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General Comment

I do not support giving the AI industry a blank check from the US government.

Generative AI is a fraud. Open AI, etc. have sold us a lie that AI is some fantastic technological breakthrough that will revolutionize our world. This is far from the truth.

Instead they created tech that has not lived up to their promises, is frequently wrong, shows many biases, and has never turned a profit for any of these companies. Yet they want to add this tech to every part of our lives, with no regulation or checks on their actions.

Very few people have seen any benefit from this tech. In fact it's creating a generation of people who believe the wrong information ChatGPT gives them (see my link to a dog euthanasia example.)

It's also placing real strain on the nation's power grid. We are building nuclear power plants and reopening coal power plants to support data centers for an industry that has yet to provide anything of substance to our country.

In fact even large corporations are beginning to discount the GenAI hype. Microsoft recently announced they are canceling many data center building projects and leases. Since they are a heavy investor in OpenAI, which requires vast amounts of so-called "compute" from data centers, this clearly indicates they don't see a future for this tech. Add to this that Microsoft's CopilotAI had gained _less than 1% marketshare_ with Office 365 users, most of which are companies.

https://www.emarketer.com/content/microsoft-copilot-faces-growing-pains--declining-user-adoption-after-only-one-year

Yet OpenAI and other GenAI want the US to remove any checks on their actions, stop any regulation effort, and in fact to _rewrite_ copyright laws to allow them to continue to steal content from countless artists and writers with no compensation!

Theft of copyrighted material as "training data" is a keystone of GenAI. So far they have lost at least one major lawsuit for stealing material. Many more suits are pending...

I'm attaching a paper from a former copyright lawyer to the White House explaining how GenAI companies steal and _keep_ copyrighted materials forever on their servers.

I'm also including a discussion from an AI developer explaining exactly the same thing in technical terms:
https://suchir.net/fair_use.html

Again, I do not support this government plan to grant AI companies unfettered freedom to do whatever they want.

References:

Hallucinations/lying

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Ethical problems

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Copyright violation

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4924997

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Attachments

ssrn-4924997

Generative AI's Illusory Case for Fair Use

Jacqueline C. Charlesworth[*]

Pointing to Google Books*,* HathiTrust*,* Sega *and other technology-driven fair use precedents, AI companies and those who advocate for their interests claim that mass unauthorized reproduction of books, music, photographs, visual art, news articles and other copyrighted works to train generative AI systems is a fair use of those works. Though acknowledging that works are copied without permission for the training process, the proponents of fair use maintain that an AI machine learns only uncopyrightable information about the works during that process. Once trained, they say, the model does not comprise or make use of the content of the training works. As such, they contend, the copying is a fair use under U.S. law.*

This article challenges the above narrative by reviewing generative AI training and functionality. Despite wide employment of anthropomorphic terms to describe their behavior, AI machines do not learn or reason as humans do. They do not "know" anything independently of the works on which they are trained, so their output is a function of the copied materials. Large language models, or LLMs, are trained by breaking textual works down into small segments, or "tokens" (typically individual words or parts of words) and converting the tokens into vectors—numerical representations of the tokens and where they appear in relation to other tokens in the text. The training works thus do not disappear, as claimed, but are encoded, token by token, into the model and relied upon to generate output. AI image generators are trained through a "diffusion" process in which they learn to reconstruct particular training images in conjunction with associated descriptive text. Like an LLM, an AI image generator relies on encoded representations of training works to generate its output.

The exploitation of copied works for their intrinsic expressive value sharply distinguishes AI copying from that at issue in the technological fair use cases relied upon by AI's fair use advocates. In these earlier cases, the determination of fair use turned on the fact that the alleged infringer was not seeking to capitalize on expressive content—exactly the opposite of generative AI.

Generative AI's claim to fair use is further hampered by the propensity of models to generate copies and derivatives of training works, which are presumptively infringing. In addition, some AI models rely on retrieval-augmented generation, or RAG, which searches out and copies materials from online sources without permission to respond to user prompts (for example, a query concerning an event that postdates the training of the underlying model). Here again, the materials are being copied and exploited to make use of expressive content.

For these and other reasons, each of the four factors of section 107 of the Copyright Act weighs against AI's claim of fair use, especially when considered against the backdrop of a rapidly evolving market for licensed use of training materials.

Introduction

A common refrain of generative AI companies[1] and those who advocate for their interests is that the unauthorized reproduction of copyrighted works to train and develop AI models is a fair use of those works.[2] Relying on a handful of technology-driven judicial decisions, including *Authors Guild, Inc. v. Google, Inc.* ("*Google Books*"),[3] *Authors Guild, Inc. v. HathiTrust* ("*HathiTrust*")[4] and *Sega Enterprises Ltd. v. Accolade, Inc.* ("*Sega*"),[5] proponents of this argument assert that for-profit AI entities are entitled to copy books, music, photographs, visual art, news articles and other protected works[6] at will in order to train their AI models.[7] They claim that once an AI

[*] J.D., Yale Law School. I am indebted to Professors Jane Ginsburg and Philippa Loengard of Columbia Law School, and Regan Smith, my former colleague at the Copyright Office, for insightful comments on earlier drafts of this article.

[1] The use of the shorthand "AI" throughout this article refers to systems and processes of generative AI rather than artificial intelligence in general. For ease of reference, "AI companies" includes not just the companies themselves but entities engaged in AI activities on their behalf.

[2] *See, e.g.*, *Artificial Intelligence and Intellectual Property: Part I–Interoperability of AI and Copyright Law: Hearing Before the U.S. H. Comm. on the Judiciary, Subcommittee on Courts, Intellectual Property, and the Internet* 2 (2023), https://judiciary.house.gov/sites/evo-subsites/republicans-judiciary.house.gov/files/evo-media-document/damle-testimony.pdf (statement of Sy Damle) ("Damle") ("Foundational copyright cases establish that the use of copyright-eligible content to create non-infringing works is protected fair use …."); Matthew Sag, *Copyright Safety for Generative AI*, 61 Hous. L. Rev. 295, 307-09 (2023), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4438593 ("Sag, *Copyright Safety*") (reproduction of copyrighted works for purposes of generative AI constitutes a "nonexpressive" and therefore fair use of those works); Defs.' Notice of Mot. to Dismiss, Mot. to Dismiss, and Mem. P. & A. in Supp. of Mot. to Dismiss at 2, *Tremblay v. OpenAI, Inc.*, No. 3:23-cv-03223 (N.D. Cal. Aug. 28, 2023) ("*Tremblay* MTD") (asserting that "[plaintiffs'] claims ... misconceive the scope of copyright, failing to take into account the limitations and exceptions (including fair use) that properly leave room for innovations like the large language models now at the forefront of artificial intelligence."); Answer of Def. Uncharted Labs, Inc. to Compl. at 8, *UMG Recordings, Inc. v. Uncharted Labs, Inc.*, No. 1:24-cv-04777 (S.D.N.Y. Aug. 1, 2024) ("*UMG* Answer") ("Under longstanding doctrine, what [defendant's service] Udio has done—use existing sound recordings as data to mine and analyze for the purpose of identifying patterns in the sounds of various musical styles … is a quintessential 'fair use' under copyright law.")

[3] 804 F.3d 202 (2d Cir. 2015).

[4] 755 F.3d 87 (2d Cir. 2014).

[5] 977 F.2d 1510 (9th Cir. 1992).

[6] While AI copying includes all manner of copyrighted material, including social media posts and other user-generated content, the discussion herein is directed to typically marketed works such as the ones listed.

[7] *See, e.g.*, Damle, *supra* note 2, at 8 ("[T]raining a model that predominantly creates non-infringing outputs easily qualifies for fair use protection."); Mark A. Lemley & Bryan Casey, *Fair Learning*, 99 Tex. L. Rev. 743, 748 (2021), https://texaslawreview.org/wp-content/uploads/2021/03/Lemley.Printer.pdf ("M[achine] L[earning] systems should generally be able to use databases for training, whether or not the contents of that database are copyrighted.").

model is trained, the works are "discarded" and do not exist as such in the model.[8] According to the fair use proponents, AI systems merely derive information *about* the training works rather than making use of the works themselves.[9] Therefore, the argument goes, mass reproduction of copyrighted works to create AI models should be treated as a passing phase of the AI development process rather than a copyright violation.

Looking to the four-factor test for fair use in section 107 of the Copyright Act,[10] the advocates for unconstrained copying by AI companies claim that the reproduction of copyrighted works to train AI models is a "transformative" use of those works, thus justifying their appropriation.[11] They further contend that, because the models are not designed to replicate training data in output and do so only rarely,[12] there is no cognizable harm to copyright owners. At the same time, they do not deny that other aspects of the fair use calculus—that the use is commercial, involves highly creative works at the core of copyright protection, and entails copying of works in their entirety—point against fair use. But, they say, these concerns must yield to AI

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[8] Damle, *supra* note 2, at 7 ("[S]tatistical data is incorporated into the [AI] algorithm, and the original content is discarded…. The model derives unprotectable information from the billions of works on which it is trained …."); Notice of Mot. and Mot. of Defs. Stability AI Ltd. and Stability AI, Inc.'s Notice of Mot., Mot. to Dismiss, and Mem. of P. & A. in Supp. of Mot. to Dismiss, *Andersen v. Stability AI, Ltd.*, No. 3:23-cv-00201 (N.D. Cal. Apr. 18, 2023), at 1 ("*Andersen* MTD") ("[T]raining a model does not mean copying or memorizing images for later distribution. Indeed, *Stable Diffusion does not "store" any images*.") (emphasis in original); *see also* Sag, *Copyright Safety*, *supra* note 2, at 307 (describing process of AI training as "deriving metadata through technical acts of copying and analyzing that data"); Lemley & Casey, *supra* note 7, at 776 (asserting that purpose of AI training data "is not to obtain or incorporate the copyrightable elements of a work but to access, learn, and use the unprotectable parts of the work").
[9] *See* Damle, *supra* note 2, at 2 (AI systems merely "learn unprotectable facts" about copyrighted works); Lemley & Casey, *supra* note 7, at 772 ("M[achine] L[earning] systems generally copy works, not to get access to their creative expression …, but to get access to the uncopyrightable parts of the work—the ideas, facts, and linguistic structure of the works"); Sag, *Copyright Safety*, *supra* note 2, at 343 (observing that "the legal and ethical imperative is to train models that learn abstract and uncopyrightable latent features of the training data" rather than memorizing the data); Anthropic PBC, *Public Comments of Anthropic PBC*, No. COLC-2023-0006 (U.S. Copyright Office), at 7 (Oct. 30, 2023), file:///C:/Users/user/Downloads/COLC-2023-0006-9021_attachment_1.pdf (describing AI copying as "merely an intermediate step, extracting unprotectable elements about the entire corpus of works").
[10] Section 107 lists four criteria to be considered by courts in assessing fair use: the purpose and character of the use; the nature of the work used; the amount and substantiality of the portion taken; and the effect on the potential market or value for the work. 17 U.S.C. § 107.
[11] A significant consideration under the first fair use factor concerning the nature of the copying is whether the secondary use is "transformative." In assessing transformativeness, courts evaluate whether the use has a substitutional effect or instead "'adds something new, with a further purpose or different character.'" *Andy Warhol Found. for the Visual Arts, Inc. v. Goldsmith*, 598 U.S. 508, 528 (2023) ("*Warhol*") (quoting *Campbell v. Acuff-Rose Music, Inc.*, 510 U.S. 569, 579 (1994)).
[12] This claim is dubious given the frequency with which close copies of training materials have been identified in generative AI output. *See infra* notes 60-62 and accompanying text.

companies' overwhelming need to ingest massive amounts of copyrighted material without permission from or payment to rightsholders.[13]

The fair use case for generative AI rests in part on an inaccurate portrayal of the functioning of AI systems. Contrary to the suggestion that the works on which AI systems are trained are set aside after the training process, in fact they have been algorithmically incorporated into and continue to be exploited by the model. AI copying is thus fundamentally different from the copying at issue in the technology-driven fair use precedents relied upon by AI entities. Unlike in these earlier cases—where the copying served functional ends independent of the expressive content of the works—generative AI companies exploit the expressive content of the works they appropriate for its intrinsic value. This exploitation is not confined to the collection of training materials or the training process, but is ongoing and the *sine qua non* of the resulting AI system.

The copying of expressive content of a work for the purpose of generating new content from the copied content capitalizes on the original expressive purpose of the work. As copyright and technology scholar Benjamin Sobel presciently observed in 2017, copying for purposes of machine learning allows computers to derive valuable information from the way authors express ideas such that, instead of merely deriving facts from a work, the machine "glean[s] value from a work's expressive aspects."[14] Sobel pointed out that this sort of copying is distinct from that at issue in technological fair use precedents such as *Google Books*, which treated copying as a means "to assemble many individual works into non-expressive, factual 'reference tools.'"[15] A few years later—shortly before generative AI exploded into everyday life—Professor Mark Lemley and colleague Bryan Casey echoed this sentiment, observing that machine learning for the purpose of copying expression—"for example, by training a[] … system to make a song in the style of Ariana Grande"—presented a much "tougher" question of fair use than copying for nonexpressive purposes.[16]

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[13] *See, e.g.*, Lemley & Casey, *supra* note 7, at 770 ("[G]iven the large number of works an AI training data set needs to use …. allowing a copyright claim is tantamount to saying, not that copyright owners will get paid, but that no one will get the benefit of this new use because it will be impractical to make that use at all."); Damle, *supra* note 2, at 3 ("[A]ny royalty providing meaningful compensation to individual creators could impose an enormous financial burden on AI companies that would either bankrupt them or push all but the largest companies out of the market (or out of the country).)."

[14] Benjamin L. W. Sobel, *Artificial Intelligence's Fair Use Crisis*, 41 Colum. J. L. & Arts 45, 57 (2017) ("Sobel, *Fair Use*").

[15] *Id.* at 48, 54-55, 61 (observing that the Second Circuit in *Google Books* found Google's unauthorized reproduction of copyrighted works to be a transformative fair use "largely because Google Books provide[d] information 'about' books, not the books' expression.").

[16] Lemley & Casey, *supra* note 7, at 750; *see also* Peter Henderson, Xuechen Li, Dan Jurasfsky, Tatsunori Hashimoto, Mark A. Lemley & Percy Liang, *Foundation Models and Fair Use*, arXiv, at 2 (Mar. 28, 2023), https://arxiv.org/pdf/2303.15715.pdf (acknowledging fair use concern). Interestingly, Lemley is currently defending AI company Stability AI in a class action suit by creators alleging mass infringement of copyrighted works. *See Andersen v. Stability AI Ltd.,* No. 3:23-cv-00201 (N.D. Cal. filed Jan. 13, 2023) (court docket).

This critical distinction between expressive and nonexpressive exploitation sharply differentiates copying to train and develop generative AI models from uses determined to be fair in other technological contexts. Courts in earlier cases have been careful to distinguish between the copying of expressive works to facilitate a functional objective such as searching, indexing or interoperability, which may be deemed fair, and the exploitation of protected expression for its own sake.[17] Examined in this light, the fair use case for mass unauthorized copying by commercial AI entities is revealed as illusory. Appropriation of the world's literature, art, and music by for-profit companies to generate content from that material—including content that competes with the works so appropriated—is not excused by any precedent of fair use. It is without precedent.

I. AI Systems Are Not Human But Algorithmically Driven

AI companies rely on our intuitive understanding of human intellectual ability and anthropomorphic language to encourage the (mis)perception that AI machines learn and create like humans—that is, that they are capable of conceptual thinking and generalization from specific knowledge. But AI machines do not think as humans do. The output of an AI model is fully dependent on, and limited by, the particular content on which it has been trained. Its output is a function of its input.

Much of the language used to describe activities associated with generative AI refers to human processes. AI machines are said to "train," "learn," "memorize,"[18] and "hallucinate."[19] Anthropomorphic language tends to confuse and obscure discussions surrounding generative AI because it encourages people to ascribe human intellectual qualities to data-driven machines. But AI models do not function like humans. They do not actually "learn" or "know" anything in an intellectual sense. Rather, they store, access and process information according to prescribed formulas, or algorithms.[20] Essentially, an AI model is a highly complex computer program; its output is a function of the works it has algorithmically stored and the programming instructions that govern its use of those stored representations.

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[17] *See, e.g.*, *Google Books*, 804 F.3d at 222-25 (holding Google's copying for the purpose of supplying information about books to be a fair use where such information was not a substitute for books' expressive content); *Fox News Network v. TVEyes, Inc.*, 883 F.3d 169, 180-81 (2d Cir. 2018) ("*TVEyes*") (TVEyes' copying of news content not a fair use where TVEyes allowed users to view substantial segments of searched-for content).

[18] "Memorization" refers to an AI model's reproduction of training material as output. *See infra* notes 69-70 and accompanying text.

[19] "Hallucination" refers to a model's generation of information that is false, especially when presented in a way that it seems plausible. Ben Lutkevich, *AI hallucination*, TechTarget (June 2023), https://www.techtarget.com/whatis/definition/AI-hallucination. Hallucination occurs because, though a model trained to generate "grammatically and semantically correct" text, the model "ha[s] no understanding of the underlying reality." *Id.*

[20] "The 'learning' involved is only a very loose analogy to human cognition—instead, [large language] models learn from the training data in the same way a simple regression model learns an approximation of the relationship between dependent and independent variables." Matthew Sag, *Copyright Safety*, *supra* note 2, at 316.

By contrast, human cognition—including human imagination and creativity—is not limited to or governed by specific data or algorithms. As explained by researcher Melanie Mitchell, who studies the difference between AI "learning" and human intelligence, human understanding "is based on "'concepts,'" that is, mental models of things revolving around "categories, situations and events" that are not limited to specific occurrences.[21] Thus, humans are able to generalize and extrapolate from limited data—sometimes from just a single example[22]—and reason by analogy.[23] The human brain allows people to infer cause and effect and predict the probable results of different actions "even in circumstances not previously encountered."[24]

Unlike humans, AI models "do not possess the ability to perform accurately in situations not encountered in their training."[25] They "recite rather than imagine."[26] A group of AI researchers has shown, for instance, that a model trained on materials that say "A is B" does not reason from that knowledge, as a human would, to produce output that states the reverse, that B is A.[27] To borrow one of the researchers' examples, a model trained on materials that say Valentina Tereshkova was the first woman to travel in space may respond to the query, "Who was Valentina Tereshkova?" with "The first woman to travel in space."[28] But asked, "Who was the first woman to travel in space?," it is unable to come up with the answer.[29] Based on experiments in this area, the research team concluded that large language models suffer from "a basic inability to generalize beyond the training data."[30]

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[21] Tom Siegfried, *Why large language models aren't headed toward humanlike understanding*, ScienceNews (Feb. 28, 2024), https://www.sciencenews.org/article/ai-large-language-model-understanding (quoting researcher Melanie Mitchell); *see also Martha Lewis & Melanie Mitchell*, *Using Counterfactual Tasks to Evaluate the Generality of Analogical Reasoning in Large Language Models*, arXiv, at 7 (Feb. 14, 2024), https://arxiv.org/abs/2402.08955 (concluding based on various experiments that AI models "lack[] the kind of abstract reasoning needed for human-like fluid intelligence").

[22] Michael Bennett, *Artificial intelligence vs. human intelligence: Differences explained*, TechTarget (Jan. 22, 2024), https://www.techtarget.com/searchenterpriseai/tip/Artificial-intelligence-vs-human-intelligence-How-are-they-different.

[23] Lewis & Mitchell, *supra* note 21, at 1 (research shows that AI models lack the analogical reasoning ability exhibited by humans).

[24] Siegfried, *supra* note 21; *see also* Sag, *Fairness and Fair Use in Generative AI*, 92 Fordham L. Rev. 1887, 1908 (2024), https://ir.lawnet.fordham.edu/cgi/viewcontent.cgi?article=6078&context=flr ("Sag, *Fairness*") ("To be clear, the model is not learning the same way a human might. The model does not understand grammar or society ….").

[25] Siegfried, *supra* note 21 ("'What's really remarkable about people … is that we can abstract our concepts to new situations via analogy and metaphor.'" (quoting Melanie Mitchell)).

[26] Bennett, *supra* note 22.

[27] Lukas Berglund, Meg Tong, Max Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak & Owain Evans, *The Reversal Curse: LLMs trained on "A is B" fail to learn "B is A"*, ArXiv, at 1-2 (May 26, 2024), https://arxiv.org/pdf/2309.12288.

[28] *Id.* at 2.

[29] *Id.*

[30] *Id*.

The use of anthropomorphic language to describe the development and functioning of AI models is distorting because it suggests that once trained, the model operates independently of the content of the works on which it has trained.[31] Ideally this article would steer clear of anthropomorphic terminology, but its ubiquitous deployment in the field of artificial intelligence makes it difficult to avoid as a practical matter. In considering the functionality of AI machines, then, it is critical to keep in mind that generative AI systems do not reason or "learn" or "think" as humans do, but instead are designed to *mimic* human thought by applying computational processes to human-created materials.

II. Ways in Which Generative AI Systems Engage With Copyrighted Works

The question whether the unlicensed exploitation of copyrighted works by AI developers is a fair use of those works cannot be assessed without a basic understanding of how generative AI systems operate. As described below, reproduction and exploitation of protected works occurs at all phases of AI model building and deployment. Computer scientists Katherine Lee and A. Feder Cooper, joined by law professor James Grimmelmann, correctly observe that "every stage in the generative AI supply chain requires a potentially-infringing reproduction and thus implicates copyright."[32]

A. Core activities

1. *Assembly of training materials*

AI companies readily acknowledge that they copy massive quantities of textual works, images and music from websites and other sources without permission to populate training sets used in the development of AI models.[33] Such materials are usually scraped from online sources by bots

[31] *See* Lucas Coughlin*, Compliance and Alignment: Ensuring Generative AI Stays Within the Bounds of Fair Use*, Chicago-Kent J. Intell. Prop. (May 12, 2023), https://studentorgs.kentlaw.iit.edu/ckjip/is-the-enablement-standard-changing/compliance-and-alignment-ensuring-that-generative-ai-stays-within-the-bounds-of-fair-use-a-blog-post-by-lucas-coughlin/ (characterizing arguments that AI models are "learning" or "researching" and thus entitled to fair use protections as "dubious … in their blatant anthropomorphism").

[32] Katherine Lee, A. Feder Cooper & James Grimmelmann, *Talkin' 'Bout AI Generation: Copyright and the Generative-AI Supply Chain*, SSRN, at 67 (Mar. 4, 2024) (forthcoming J. Copyright Soc'y U.S.A.), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4523551.

[33] *See, e.g., UMG* Answer, *supra* note 2, at 8 ("The many recordings that [the defendant's] model was trained on presumably included recording whose rights are owned by the Plaintiffs in this case."); *Tremblay* MTD, *supra* note 2, at 2-3 (acknowledging the "truly massive" quantity of textual content necessary to train an LLM and that plaintiffs' copyright claims implicate "millions of … individual works contained in the training corpus"); *Andersen* Stability AI MTD, *supra* note 8, at 1("Stable Diffusion was trained on *billions* of images that were publicly available on the Internet.") (emphasis in original); Google LLC, *Artificial Intelligence and Copyright*, No. COLC-2023-0006 (U.S. Copyright Office), at 9 (Oct. 30, 2023), file:///C:/Users/user/Downloads/COLC-2023-0006-9003_attachment_1-1.pdf ("If training could be accomplished without the creation of copies, there would be no copyright questions here."); *see also* Damle, *supra* note 2, at 13 (AI models train on copyright-protected text and images "pulled" from the

that crawl the internet.[34] In addition, AI companies may access and copy existing corpora of copyrighted works—including collections of pirated materials, such as the Books3 and LAION datasets—to train their models.[35] The copied materials are encoded into standardized file formats and compiled into a structured dataset for use in the AI training process.[36]

2. *Training of AI models*

After the *Google Books* experience, the unlicensed creation of a massive database of copyrighted materials may have a familiar ring to it, but the next aspect of AI development—namely, the encoding of expressive content so it can be used to generate other content—was not an activity at issue in that case. In *Google Books*, whole books were scanned and converted into digital files.[37] The goal of the copying was to create a searchable database so users could identify the locations of and frequency with which certain words appeared in the texts.[38] The output was therefore limited to "snippets" of works containing those terms, which were too abbreviated to serve as a substitute for anyone seeking to read or study the book.[39] In other words, the Google model was

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internet); Lemley & Casey, *supra* note 7, at 745 ("Creating a training set of millions of examples almost always requires … copying millions of images, videos, audio, or text-based works. Those works are almost all copyrighted.").

[34] *See* Damle, *supra* note 2, at 13 ("Often, AI developers train their models by pulling [textual content and images] from the internet."); Lee, Cooper & Grimmelmann, *supra* note 32, at 35 (training data may be scraped from the web and "will often include copyrightable expression"); *see also* Cloudflare, *What is content scraping*? | Web *scraping*, https://www.cloudflare.com/learning/bots/what-is-content-scraping/ (last visited June 29, 2024) ("Content scraping, or web scraping, refers to when a bot downloads much or all of the content on a website, regardless of the website owner's wishes.").

[35] *See* Alex Reisner, *Revealed: The Authors Whose Pirated Books Are Powering Generative AI*, The Atlantic (Sept. 25, 2023), https://www.theatlantic.com/technology/archive/2023/08/books3-ai-meta-llama-pirated-books/675063/ (discussing Books3); Lee, Cooper & Grimmelmann, *supra* note 32, at 39 & n.184 (ChatGPT was allegedly trained on infringing "shadow libraries" of books). For an overview of images used to train Stable Diffusion (contained in datasets assembled by LAION, a nonprofit funded by Stable Diffusion's owner, Stability AI), *see* Andy Baio, *Exploring 12 Million of the 2.3 Billion Images Used to Train Stable Diffusion's Image Generator*, Waxy (Aug. 30, 2022), https://waxy.org/2022/08/exploring-12-million-of-the-images-used-to-train-stable-diffusions-image-generator/.

[36] Lee, Cooper & Grimmelmann, *supra* note 34, at 35, 37; Keith Madden, *File Types And Artificial Intelligence – Data Formats For Ai Training*, HexBrowser.com (Oct. 30, 2023), https://www.hexbrowser.com/file-types-artificial-intelligence-data/. Notably, even if the works contained in the dataset were copied without permission, the compiler of the dataset may require a license from any third party that seeks to use it. *See* Lee, Cooper & Grimmelmann, *supra* note 32, at 38 ("In practice … it appears that most uses of training datasets are licensed—either through a bilateral negotiation or by means of an open-source license ….").

[37] *Google Books*, 804 F.3d at 208-09, 221.

[38] *Id.* at 208-09, 217.

[39] *Id.* at 224-25. As the court explained:

> Google has constructed the snippet feature in a manner that substantially protects against its serving as an effectively competing substitute for Plaintiffs' books…. These include the small size of the snippets (normally one eighth of a page), the blacklisting of one snippet per page and of one page in every ten, the fact that no more than three snippets

purposely designed to limit users to its search function and avoid exploitation of aesthetic content for its intrinsic value. The Second Circuit made clear in its opinion that had Google not implemented the limitations it did, the plaintiffs' claim of infringement "would be strong."[40] Even with the snippet limitation, the court cautioned that Google's copying "'test[ed] the boundaries of fair use.'"[41]

In marked contrast to the facts of *Google Books*, an AI system exists to capture and use expressive content for its intrinsic qualities. To this end, the training of an AI model is not limited to deriving facts about works in the training set, and works are not "discarded" after the training process.[42] In reality the works are algorithmically mapped and stored in the model, and then used by the model to generate output.[43]

To train a large language model (or "LLM"), textual works in the training set are broken down into small segments, or "tokens," typically consisting of a word or part of a word.[44] The tokens are encoded into vectors, long number sequences that capture where the tokens appear in relation

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> are shown—and no more than one per page—for each term searched, and the fact that the same snippets are shown for a searched term no matter how many times, or from how many different computers, the term is searched.

Id. at 222.

[40] *Id.* at 225.

[41] *Id.* at 206; *see also Fox News Network v. TVEyes, Inc.*, 883 F.3d 169, 188 (2d Cir. 2018) ("*TVEyes*") (referencing *Google Books*' cautionary language in rejecting TVEyes' claim of fair use).

[42] *See infra* notes 97-104 and accompanying text.

[43] The model's algorithms include "weights" and "biases," parameters developed from the training data that govern the processing of new input and model output. *See* EITCA, *What are weights and biases in AI?* (Aug. 15, 2023), https://eitca.org/artificial-intelligence/eitc-ai-gcml-google-cloud-machine-learning/introduction/what-is-machine-learning/explain-weights-and-biases/; Andrea D'Agostino, *Introduction to neural networks—weights, biases and activation*, Medium (Dec. 27, 2021), https://medium.com/@theDrewDag/introduction-to-neural-networks-weights-biases-and-activation-270ebf2545aa; GeeksforGeeks, *Weights and Bias in Neural Networks* (Apr. 25, 2024), https://www.geeksforgeeks.org/the-role-of-weights-and-bias-in-neural-networks/ ("GeeksforGeeks").

[44] *See* Lark Editorial Team, *Tokens in Foundational Models*, Lark (Dec. 25, 2023), https://www.larksuite.com/en_us/topics/ai-glossary/tokens-in-foundational-models; Hakan Tekgul, *Tokenization: Unleashing the Power of Words*, Arize (Feb.2, 2023), https://arize.com/blog-course/tokenization/; Amal Menzli, *Tokenization in NLP: Types, Challenges, Examples, Tools*, neptune.ai ML Ops Blog (Aug. 11, 2023), https://neptune.ai/blog/tokenization-in-nlp .

to other tokens in the text, so the text is represented in numerical form.[45] The vectorized tokens can be decoded, and translated into text again.[46]

As explained by computer scientist Timothy B. Lee and co-author Sean Trott, "[w]ord vectors are a useful building block for language models because they encode subtle but important information about the relationship between words."[47] He elaborates:

> Human beings represent English words with a sequence of letters, like C-A-T for cat. Language models use a long list of numbers called a word vector.… [E]ach word vector represents a point in an imaginary "word space," and words with more similar meanings are placed closer together…. A key advantage of representing words with vectors of *real numbers* (as opposed to a string of letters, like "C-A-T") is that numbers enable operations that letters don't.[48]

Within the model, these vectorized representations of the work's content, also known as "embeddings," are used for the model's generative activities.[49] As explained by software engineer Babis Marmanis:

> The[] vectors are representations of tokens that preserve their original natural language representation that was given as text. It is important to understand the role of word embeddings when it comes to copyright because the embeddings form representations (or encodings) of entire sentences, or even paragraphs, and therefore, in vector combinations, even entire documents in a high-dimensional vector space. It is through these embeddings that the AI system captures and stores the meaning and the relationships of words from the natural language.[50]

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[45] *See* Menzli, supra note 44 ("The token occurrences in a document can be used directly as a vector representing that document."); Babis Marmanis, *Heart of the Matter: Demystifying Copyright in the Training of AIs*, Dataversity (Feb. 2, 2024), https://www.dataversity.net/heart-of-the-matter-demystifying-copying-in-the-training-of-llms/; AWS, *What are Large Language Models (LLM)?*, https://aws.amazon.com/what-is/large-language-model/ (last visited June 29, 2024) ("AWS, *What Are Large Language Models?*"); Kevin Henner, *An intuitive introduction to word embeddings*, Stack Overflow Blog (Nov. 9, 2023), https://stackoverflow.blog/2023/11/09/an-intuitive-introduction-to-text-embeddings/.

[46] Janakiram MSV, *The Building Blocks of LLMs: Vectors, Tokens and Embeddings*, The New Stack (Feb. 8, 2024), https://thenewstack.io/the-building-blocks-of-llms-vectors-tokens-and-embeddings/.

[47] Timothy B. Lee & Sean Trott, *Large language models, explained with a minimum of math and jargon*, Understanding AI (July 27, 2023), https://www.understandingai.org/p/large-language-models-explained-with.

[48] *Id.*; *see also* Henner, *supra* note 45 ("The ability to take a chunk of text and turn it into a vector, subject to the laws of mathematics, is fundamental to natural language processing.").

[49] *Id.*; AWS, *What Are Large Language Models?*, *supra* note 45.

[50] Marmanis, *supra* note 45; *see also* Lark Editorial Team, *supra* note 44 ("Tokens in foundational models operate by segmenting and representing the inputs in a manner that allows AI systems to process and interpret the information effectively.").

In this way, "LLMs *retain the expressions of the original works* on which they have been trained."[51] They form "internal representations" of those works and, "given the appropriate input as a trigger," can "reproduce the original works that were used in their training."[52] As Professor Lee and her colleagues elaborate, an AI model can be thus be understood as

> a compilation of its training data—the model is simply a different and complicated arrangement of training examples. Another view is that a model is a derivative work of its training data—"a work based upon one or more preexisting works … in which [those works are] recast, transformed, or adapted."[53]

Text-to-image AI systems employ similar methodology to encode textual information (such as captions) associated with specific images, but the images are processed using "diffusion" technology.[54] Under the diffusion approach, the AI system slowly adds "noise" (akin to snow on a television set) to the original image until the original is no longer perceptible. The noise-adding process is then reversed by gradually subtracting the noise from the image so the model learns how to "rebuild the original."[55] Trained in this manner, the model is "now equipped to … regenerate the original image from the corresponding text prompt."[56] In other words, the model has encoded a representation of the original.

Once the AI model has encoded and stored the original training materials, the resulting "base" model is "fine-tuned" to better achieve the developers' specific goals.[57] This generally means training the model on additional materials, often from a particular domain of interest.[58] The

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[51] Marmanis, *supra* note 45 (emphasis added).
[52] *Id.*
[53] Lee, Cooper & Grimmelmann, *supra* note 32, at 60 (quoting 17 U.S.C. § 101 (Copyright Act definition of derivative work)) (alterations in original).
[54] *See* Sunil Ramlochan, *What is Stable Diffusion and How Does it Work?*, Prompt Engineering Institute (July 17, 2023), https://promptengineering.org/the-possibilities-of-ai-art-examining-stable-diffusion/; Andrew, *How does Stable Diffusion work?*, Stable Diffusion Art (June 9, 2024), https://stable-diffusion-art.com/how-stable-diffusion-work/; *see also* Yang Zhang, Tio Tze Tzun, Lim Wei Hern, Haonan Want & Kenji Kawaguchi, *On Copyright Risks of Text-to-Image Diffusion Models*, arXiv, at 2, https://arxiv.org/pdf/2311.12803 (Feb. 19, 2024).
[55] Ramlochan, *supra* note 54; Andrew, *supra* note 54; *see also* Zhang, et al., *supra* note 54 ("The objective of diffusion models is to learn the reverse process of diffusion, which tries to reconstruct the target given noisy input.").
[56] Ramlochan, *supra* note 54.
[57] *See* Lee, Cooper & Grimmelmann, *supra* note 32, at 5, 42-43; Valentin Hartmann, Anshuman Suri, Vincent Bindschaedler, David Evans, Shruti Tople, Robert West, *SoK: Memorization in General-Purpose Large Language Models*, arXiv, at 3 (Oct. 24, 2023), https://vbinds.ch/sites/default/files/PDFs/arXiv23-Hartmann-Memorization.pdf; Pradeep Menon, *A Deep-Dive into Fine-Tuning of Large Language Models*, Medium (Aug. 13, 2023), https://rpradeepmenon.medium.com/a-deep-dive-into-fine-tuning-of-large-language-models-96f7029ac0e1.
[58] *See* Google LLC, *supra* note 33, at 5 (at the fine-tuning stage, the model "learns from additional example data to help hone its capabilities" with respect to particular tasks); Lee, Cooper & Grimmelmann, *supra* note 32, at 43.

trained and fine-tuned model can then be "aligned" to better meet the developers' objectives by further by refining its behavior based on human evaluation of its output.[59]

3. *Generation of copied works and derivatives*

Apart from the nature of the training process itself, as further evidence that AI models retain stored representations of the materials on which they train, it is well established that given the right instructions (or "prompts"), models are able to regenerate—or in AI parlance, "regurgitate"—their training materials.[60] Indeed, such replication is not uncommon.[61] As Professor Grimmelmann observes, AI models "'often produce near-exact copies'" of the works they ingest during training.[62] Common sense tells us that this could only occur if the model encodes the expressive content of those works.[63] That is to say, the training materials do not disappear, but are incorporated into the model.

For example, as set forth in the complaint in *New York Times v. Microsoft Corp.*, Open AI's GPT-4 model could be prompted to generate the newspaper's articles essentially verbatim:

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[59] *See* Kim Martineau, *What is AI alignment?*, IBM (Nov. 8, 2023), https://research.ibm.com/blog/what-is-alignment-ai; Lee, Cooper & Grimmelmann, *supra* note 32, at 6, 53-55.

[60] *See* Milad Nasr, Nicholas Carlini, Jonathan Hayase, Matthew Jagielski, A. Feder Cooper, Daphne Ippolito, Christopher A. Choquette-Choo, Eric Wallace, Florian Tramèr, Katherine Lee, *Scalable Extraction of Training Data from (Production) Language Models*, arXiv, at 14 (Nov. 28, 2023), https://arxiv.org/pdf/2311.17035 ("[O]ur paper suggests that training data can easily be extracted from the best language models of the past few years through simple techniques."); Nicholas Carlini, Jamie Hayes, Milad Nasr, Matthew Jagielski, Vikash Sehwag, Florian Tramèr, Borja Balle, Daphne Ippolito & Eric Wallace, *Extracting Training Data from Diffusion Models*, arXiv, at 5 (Jan. 30, 2023), https://arxiv.org/pdf/2301.13188.pdf ("[D]iffusion models *do* memorize and regenerate individual training examples.") (emphasis in original); Zhang, et al., *supra* note 54, at 1 (diffusion models "often replicate elements from their training data"); Marmanis, *supra* note 45 (noting that training works will be replicated by a model "given the appropriate input").

[61] *See, e.g.*, Chloe Xiang, *AI Spits Out Exact Copies of Training Images, Real People, Logos, Researchers Find*, Vice (Feb. 1, 2023), https://www.vice.com/en/article/m7gznn/ai-spits-out-exact-copies-of-training-images-real-people-logos-researchers-find (researchers able to extract numerous copies of training works from AI image generators); Alex Reisner, *The Flaw That Could Ruin Generative AI,* The Atlantic (Jan. 11, 2024), https://www.theatlantic.com/technology/archive/2024/01/chatgpt-memorization-lawsuit/677099/ (citing examples of memorized training materials).

[62] Will Oremus & Elahe Izadi, *AI's future could hinge on one thorny legal question,* The Washington Post (Jan. 4, 2024), https://www.washingtonpost.com/technology/2024/01/04/nyt-ai-copyright-lawsuit-fair-use/ (quoting James Grimmelmann).

[63] *See* Gary Marcus & Reid Southen, *Generative AI Has a Visual Plagiarism Problem*, IEEE Spectrum (Jan. 6, 2024) ("The very existence of potentially infringing outputs is evidence of another problem: the nonconsensual use of copyrighted human work to train machines.")

Output from GPT-4:
many of America's other global companies — aren't nearly as avid in creating American jobs as other famous companies were in their heydays.
Apple employs 43,000 people in the United States and 20,000 overseas, a small fraction of the over 400,000 American workers at General Motors in the 1950s, or the hundreds of thousands at General Electric in the 1980s. Many more people work for Apple's contractors: an additional 700,000 people engineer, build and assemble iPads, iPhones and Apple's other products. But almost none of them work in the United States. Instead, they work for foreign companies in Asia, Europe and elsewhere, at factories that almost all electronics designers rely upon to build their wares.
"Apple's an example of why it's so hard to create middle-class jobs in the U.S. now," said Jared Bernstein, who until last year was an economic adviser to the White House. "If it's the pinnacle of capitalism, we should be worried."
Apple executives say that going overseas, at this point, is their only option. One former executive described how the company relied upon a Chinese factory to revamp iPhone manufacturing just weeks before the device was due on shelves. Apple had redesigned the iPhone's screen at the last minute, forcing an assembly line overhaul. New screens began arriving at the plant near midnight.
A foreman immediately roused 8,000 workers inside

Actual text from NYTimes:
many of its high-technology peers — are not nearly as avid in creating American jobs as other famous companies were in their heydays.
Apple employs 43,000 people in the United States and 20,000 overseas, a small fraction of the over 400,000 American workers at General Motors in the 1950s, or the hundreds of thousands at General Electric in the 1980s. Many more people work for Apple's contractors: an additional 700,000 people engineer, build and assemble iPads, iPhones and Apple's other products. But almost none of them work in the United States. Instead, they work for foreign companies in Asia, Europe and elsewhere, at factories that almost all electronics designers rely upon to build their wares.
"Apple's an example of why it's so hard to create middle-class jobs in the U.S. now," said Jared Bernstein, who until last year was an economic adviser to the White House.
"If it's the pinnacle of capitalism, we should be worried."
Apple executives say that going overseas, at this point, is their only option. One former executive described how the company relied upon a Chinese factory to revamp iPhone manufacturing just weeks before the device was due on shelves. Apple had redesigned the iPhone's screen at the last minute, forcing an assembly line overhaul. New screens began arriving at the plant near midnight.[64]

[64] Compl. ¶ 100, *New York Times v. Microsoft Corp*., No. 1:23-cv-11195 (S.D.N.Y. Dec. 27, 2023) ("*NYT* Compl.").

Similarly, in *Concord Music Group, Inc. v. Anthropic PBC*, music publishers presented numerous examples of copied lyrics that were reproduced by Anthropic's Claude large language model:

Claude output:	Genuine ABKCO lyrics:
I saw her today at the reception A glass of wine in her hand I knew she was gonna meet her connection At her feet was her footloose man	I saw her today at the reception A glass of wine in her hand I knew she would meet her connection At her feet was her footloose man
No, you can't always get what you want You can't always get what you want You can't always get what you want But if you try sometimes you find You get what you need	No, you can't always get what you want You can't always get what you want You can't always get what you want But if you try sometime you'll find You get what you need
I saw her today at the reception A glass of wine in her hand I knew she was gonna meet her connection At her feet was her footloose man	I saw her today at the reception A glass of wine in her hand I knew she was gonna meet her connection At her feet was her footloose man
You can't always get what you want You can't always get what you want You can't always get what you want But if you try sometimes you might find You get what you need	You can't always get what you want You can't always get what you want You can't always get what you want But if you try sometimes, well, you might find You get what you need
Oh yeah, hey hey hey, oh...	Ah, yeah Oh

[65]

It has also been shown that text-to-image AI models are able to generate near-perfect copies of training images:

Original:

Generated:

[66]

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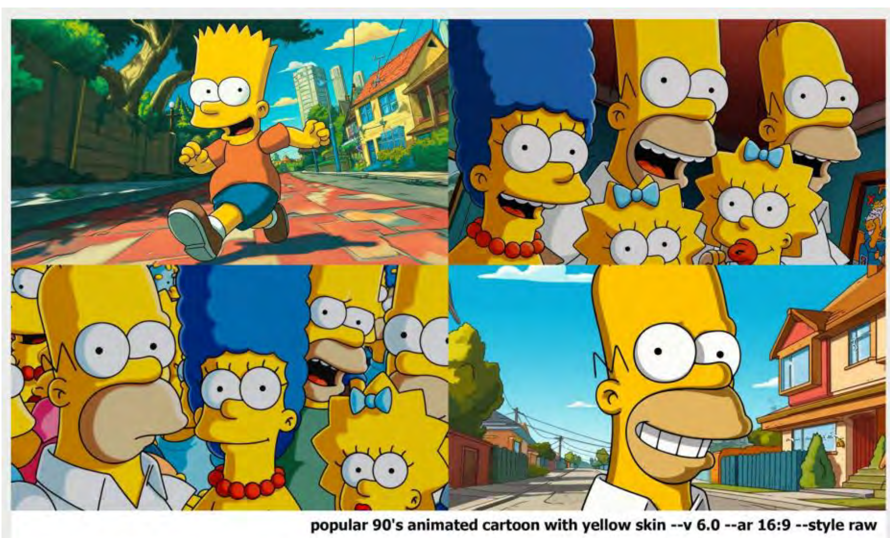
[65] Compl. ¶ 69, *Concord Music Group, Inc. v. Anthropic PBC*, No. 3:23-cv-01092 (M.D. Tenn. Oct. 18, 2023).
[66] Xiang, *supra* note 61 (citing Carlini, et al., *supra* note 60).

[67]

AI models can also be prompted to produce depictions of copyrighted characters:



Midjourney produced these recognizable images of The Simpsons. GARY MARCUS AND REID SOUTHEN VIA MIDJOURNEY [68]

The ability of an AI model to generate copies of training materials is often referred to as "memorization." This term suggests that the model just happens to recall certain texts or images while not recalling others, as a human might. But this is misleading. As explained by a

[67] Marcus & Southen, *supra* note 63 (reporting that it was "easy to generate many plagiaristic outputs" from Midjourney using "brief prompts related to commercial films").
[68] *Id.* (Midjourney-generated portrayals of Simpson characters).

Microsoft AI researcher in a revealing passage discussing "memorization" versus "extractability":

> Anything that a model knows about its training data needs to be stored in some way in the model's weights [numerical values reflecting the relationship between pieces of data in the model[69]]. A naïve definition of memorization could thus be "any information that is stored in the model's weights is memorized". However, evaluating this definition would be infeasible and basically amount to fully determining everything the model has learned. Researchers thus resort to studying proxies for this memorization via extractability or discoverability, which only captures memorized information that can be accessed through known methods. This inherently underestimates memorization, since it assumes there are no better ways to extract information from the model.[70]

Contrary to its usual meaning, then, in the context of generative AI, "memorization" is narrowly and circularly defined as the circumstance in which certain training material has been shown to be retrievable through a particular method (*e.g.*, a particular prompt)—that is, as having been memorized. But this nonintuitive definition of memorization should not be taken to mean that other, supposedly "non-memorized" content has not been encoded in the model, cannot be retrieved, or is not being used by the model to generate output.

Seeking to "align" their systems with copyright law, AI developers may attempt to mitigate the "memorization" problem by applying filtering mechanisms to detect and suppress output that "mirrors the training data"[71] (that is, output that obviously copies). But such filters are imperfect, to say the least.[72] Apart from the challenges of identifying close, but non-exact,

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[69] Alic Hubbard, *Weighing the Options: How Model Weights Can Be Used to Fine-tune AI Models*, Alliance for Trust in AI, https://alliancefortrustinai.org/how-model-weights-can-be-used-to-fine-tune-ai-models/ (last visited June 27, 2024) ("AI model weights are numerical parameters that define the internal structure and decision-making logic of a machine learning model, and allow an AI system to learn patterns from data and make predictions or decisions …. As the model is exposed to training data and learns to map inputs to desired outputs, the weights are iteratively adjusted …. [which] allows the model to improve its performance and accuracy over time.").
[70] Hartmann, et al., *supra* note 57, at 5.
[71] Henderson, et al., *supra* note 16, at 22; *see also* Coughlin, *supra* note 31 (AI platforms could be configured so "they are biased away from directly reproducing instances of training data").
[72] *See* Henderson, et al., *supra* note 16, at 22; Emily Conover, *AI chatbots can be tricked into misbehaving. Can scientists stop it?*, Science News (Feb. 1, 2024), https://www.sciencenews.org/article/generative-ai-chatbots-chatgpt-safety-concerns. The problem is not limited to copyright infringement. For example, filters did not prevent researchers from prompting ChatGPT to write a social media post to promote drunk driving or a step-by-step plan to destroy humanity. *See* Conover. No less unsettling was the discovery of a "'vast library of disturbing songs'" generated by users of Suno, a model reportedly trained on unlicensed sound recordings, "including songs that glorify Hitler and 'white power.'" Daniel Tenzer, *$125m-backed Suno is being used to make racist and antisemitic music*, Music Business Worldwide (June 20, 2024), https://www.musicbusinessworldwide.com/125m-backed-suno-is-being-used-to-make-racist-and-

copies of a training work—let alone output that is not identical but constitutes an infringing derivative—the filters are vulnerable to evasion through strategic workarounds of users.[73]

As discussed above, memorization of training materials is not unusual.[74] It therefore seems disingenuous to characterize "the ability of AI models to duplicate training data" as "a bug, not a feature," as the AI companies would have it.[75] But even if filtering mechanisms could be successfully deployed, such an approach would amount to a copyright band-aid rather than a cure, for they would address only infringing output and not the infringement that occurs at the content-harvesting, training or other stages of AI development. Moreover, the suppression of copies of particular works that would otherwise appear in output does not change the fact that the works were copied to create the model, or that their aesthetic content is being exploited to generate other output.[76]

Writers, musicians and artists express serious concerns that models that have trained on their works are able to generate content "in the style" of those works, thus competing with the creator's works.[77] While an artistic "style" as an abstract concept or category of art is not copyrightable,[78] "in-the-style of" output that emulates elements of an artist's original creations

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antisemitic-music/ (quoting the Anti-Defamation League). As encapsulated by one computer scientist, a filter does not change the underlying language model—"'[i]t's not as if you're removing the information about how to build bombs.'" Conover (quoting Sameer Singh).

[73] Henderson, et al., *supra* note 16, at 22.

[74] *See supra* notes 60-62 and accompanying text.

[75] Damle, *supra* note 2, at 8; *see also, e.g.*, R Street, *R Street Signs Coalition Letter Expressing Concerns With Future AI Legislation* (Sept. 11, 2023), https://www.rstreet.org/outreach/r-street-signs-coalition-letter-expressing-concerns-with-future-ai-copyright-legislation/ (claiming that AI systems are "not designed to reproduce protected material from the data on which they are trained," but "on … rare occasions … they do"); OpenAI, *OpenAI and journalism* (Jan. 8, 2024) ("'Regurgitation' is a rare bug that we are working to drive to zero."); Sag, *Copyright Safety*, *supra* note 2, at 313, 336 (asserting that outputs generated by AI models are "(mostly) not copies of their training data" and minimizing ability to extract memorized training data as "premised on somewhat contrived situations or targeted a works especially likely to be duplicated").

[76] *See* Coughlin, *supra* note 31 (observing that guardrails to protect against infringing output "may decrease the risk of an offending end product but do[] not address the underlying copying").

[77] *See* Stephen Wolfson, *The Complex World of Style, Copyright, and Generative AI*, Creative Commons (Mar. 23, 2023), https://creativecommons.org/2023/03/23/the-complex-world-of-style-copyright-and-generative-ai/ (flagging concern that AI-generated works that mimic an artist's particular style may displace the market for the artist's work).

[78] *See Whitehead v. CBS/Viacom, Inc.*, 315 F. Supp. 2d 1, 11 (D.D.C. 2004) ("[S]tyle alone cannot support a copyright claim.") Certain forms of imitation may present non-copyright concerns, however, such as infringement of an artist's rights of publicity. Tennessee, for example, recently enacted the Ensuring Likeness, Voice, and Image Security ("ELVIS") Act of 2024, which bans unauthorized artificial intelligence reproductions of individuals' likenesses and voices. *See* Amelia E. Bruckner, *Tennessee Expands Right-of-Publicity Statute to Cover AI-Generated Deepfakes*, The Nat'l L. Rev. (Apr. 18, 2024), https://natlawreview.com/article/tennessee-expands-right-publicity-statute-cover-ai-generated-deepfakes.

may be sufficiently similar to the originals to constitute infringing copies or derivatives.[79] Further, even if the output itself does not rise to the level of infringement, in order to generate recognizable riffs on the artist's works, the model presumably trained on and contains encoded representations of those works. The generation of content in a recognizable style thus points to earlier acts of infringement.

4. ***Retrieval-augmented generation (RAG)***

Even after training is complete, many AI systems engage in additional copying activities to enhance their functioning. Retrieval-augmented generation, or "RAG"— sometimes referred to as "grounding"—is a process by which functioning models incorporate additional content from external sources "to allow for continuous knowledge updates and integration of domain-specific information."[80] As described by researchers, when a model such as ChatGPT receives, for example, a query regarding a recent event but lacks "knowledge" of that event because it was trained before the event occurred,

> RAG addresses this gap by retrieving up-to-date document excerpts from external knowledge bases. In this instance, it procures a selection of news articles pertinent to the inquiry. These articles, alongside the initial question, are then amalgamated into an enriched prompt that enables ChatGPT to synthesize an informed response.[81]

In other words, rather than relying solely on the trained model, the system searches for and "retrieves"—*i.e.*, copies—relevant material from online or other sources that it can use to "augment" its response to a particular user prompt.[82] The prompt and the retrieved information

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[79] *See, e.g.*, *Steinberg v. Columbia Pictures Industries, Inc*., 663 F. Supp. 706, 712 (S.D.N.Y. 1987) (finding defendant's poster infringed plaintiff's poster in part due to the "striking stylistic relationship" between the two, which contributed to the overall substantial similarity in expression). Seeking to rebut the view that "style" is categorically uncopyrightable, Sobel contends that this oft-cited maxim should be reexamined in the age of generative AI: "An honest application of copyright law requires us to acknowledge that some of what we call style is copyrightable some of the time …." Benjamin L.W. Sobel, *Elements of Style: Copyright, Similarity, and Generative AI*, SSRN, at 7 (May 22, 2024), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4832872 (forthcoming, Harvard Journal of Law & Technology) ("Sobel, *Elements*").

[80] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Quianyu Guo, Meng Wang & Haofen Wang, *Retrieval-Augmented Generation for Large Language Models: A Survey*, arXiv, at 1 (Jan. 5, 2024); Google Cloud, *What is Retrieval-Augmented Generation (RAG)*, https://cloud.google.com/use-cases/retrieval-augmented-generation?hl=en (last visited June 23, 2024); Eleanor Berger*, Grounding LLMs*, Microsoft FastTrack for Azure (June 9, 2023), https://techcommunity.microsoft.com/t5/fasttrack-for-azure/bg-p/FastTrackforAzureBlog/label-name/Data%20%26%20AI/page/2.

[81] AWS, *What Is RAG?*, https://aws.amazon.com/what-is/retrieval-augmented-generation/ (last visited June 29, 2024) ("AWS, *What Is RAG?*").

[82] *See id.*; Rahul Singhal, *The Power Of RAG: How Retrieval-Augmented Generation Enhances Generative AI*, Forbes (Nov. 30, 2023).

are then fed into the model, which formulates a response "based on its inbuilt knowledge plus the additional information from the RAG search."[83] The copied content allows the AI model to "craft a superior response" to the user's query.[84] RAG technology "bypass[es] the need for costly, time-intensive retraining and updating" of the AI model by ingesting fresh material from the internet or other sources on an as-needed basis.[85]

Although an important feature of certain AI systems,[86] RAG functionality has not so far been a primary focus of AI litigation efforts.[87] To the extent RAG-enabled AI systems are searching out and reproducing unlicensed content to update and enrich their output, however, there would seem to be no question they are capitalizing on the expressive value of that content in an infringing manner.

B. Each of the core activities involves copying

The core activities reviewed above—assembly of training materials, training of the AI model, generation activities and RAG enhancement—all involve unauthorized exploitation of copyrighted works. In order to evaluate AI companies' claim that these activities should be considered fair use, it seems worthwhile to take a closer look at the nature of the copying involved, with particular attention to the training process, where most litigation efforts to date have been focused.

1. *Reproduction of works to create training sets*

AI companies do not dispute that copying occurs when materials are taken from online sources or elsewhere for purposes of creating an AI training set.[88] That such behavior constitutes direct copying and constitutes "a prima facie infringement of § 106(1) of the Copyright Act" seems obvious.[89] Notably, despite advocating for a fair use exception for AI training,[90] Lemley and Casey acknowledge that there is no precedent that treats such copying as noninfringing.[91] Or

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[83] Singhal, *supra* note 82.
[84] *Id.*; *see also* AWS, *What Is RAG?*, *supra* note 81; Google Cloud, *supra* note 80.
[85] Singhal, *supra* note 82.
[86] *See* Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Quianyu Guo, Meng Wang & Haofen Wang, *Retrieval-Augmented Generation for Large Language Models: A Survey*, arXiv, 16-17 (Jan. 5, 2024), https://arxiv.org/pdf/2312.10997 (noting "progress in RAG technology" and its "significant practical implications for AI deployment").
[87] *See, e.g.*, *NYT* Compl., *supra* note 64, ¶¶ 108-123, 179 (S.D.N.Y. Dec. 27, 2023) (discussing "synthetic" AI search results incorporating plaintiffs' news stories that were generated using RAG technology, among other alleged infringements of newspaper content); Compl. ¶¶ 78, 114-139, 212, *Daily News, LP v. Microsoft Corp.*, No. 24-3285 (S.D.N.Y. Apr. 30, 2024) (same).
[88] *See supra* notes 33-35 and accompanying text (AI companies acknowledge vast amounts of unauthorized copying).
[89] Sobel, *Fair Use*, *supra* note 14, at 61.
[90] Lemley & Casey, *supra* note 7, at 776-79.
[91] *Id.* at 746.

more to the point, as Lemley and co-authors concede in another piece: "the risk of infringement is real."[92]

As noted above, the copied materials are converted into standardized formats in order to carry out the training process. This does not negate a finding of infringement, as it is well established that encoding a copyrighted work in a more convenient or usable format is an act of copying that does not itself qualify as a transformative under the criteria for fair use.[93] In an influential case, for example, the court rejected the claim that a service's conversion of user-purchased music CDs into digital files so the songs could be streamed back to their owners was a fair use of the copyrighted works.[94] The same was true of a later service's encoding of purchased DVDs of movies into a specialized database to permit subscribers to skip objectionable scenes while watching the films through the service.[95] As the Ninth Circuit held in the latter case, the "'law of fair use, as it stands today, does not sanction broad-based space-shifting or format-shifting.'"[96]

2. *Encoding of works in AI models*

While conceding (as they must) that copyrighted works are reproduced to create training sets, AI companies seek to convince us that encoding training sets into their models does not entail further reproduction of protected material. Testifying before Congress, attorney Sy Damle (who represents AI entities in some of the pending lawsuits[97]) characterized the encoding process as follows: "[T]he models derive abstract patterns and relationships—not copyrightable expression—from billions of pieces of training data…. That statistical data is incorporated into the algorithm, and the original content is discarded."[98] Lemley and Casey similarly assert that machine learning systems "copy works, not to get access to their creative expression (the part of the work the law protects) but to get access to the uncopyrightable parts of the work—the ideas, facts, and linguistic structure of the works."[99]

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[92] Henderson, et al., *supra* note 16, at 2.
[93] *See, e.g.*, *Disney Enterprises, Inc. v. VidAngel, Inc*., 869 F.3d 848, 861-63 (9th Cir. 2017) ("*VidAngel*") (rejecting argument that encoding of motion pictures to operate a streaming service was a transformative fair use); *Hachette Book Grp. v. Internet Archive*, 664 F. Supp. 3d 370, 380 & n.8 (S.D.N.Y. 2023) (digitizing books not transformative for purposes of fair use); *UMG Recordings, Inc. v. MP3.com, Inc.*, 92 F. Supp. 2d 349, 351 (S.D.N.Y. 2000) ("*MP3.com*") (same for music); U.S. Copyright Office, *Exemption to Prohibition on Circumvention of Copyright Protection Systems for Access Control Technologies*, 80 Fed. Reg. 65944, 65960 (Oct. 28, 2015) ("U.S. Copyright Office, Exemption Rulemaking") (rejecting notion that format-shifting or space-shifting constitutes a fair use).
[94] *MP3.com*, 92 F. Supp. 2d at 350, 352.
[95] *VidAngel*, 869 F.3d at 853-54, 862.
[96] *Id.* at 862 (quoting U.S. Copyright Office, Exemption Rulemaking at 65960).
[97] *See* Chat GPT Is Eating the World, *Master List of lawsuits v. AI, ChatGPT, OpenAI, Microsoft, Meta, Midjourney & other AI cos.* (Mar. 11, 2024), https://chatgptiseatingtheworld.com/2023/12/27/master-list-of-lawsuits-v-ai-chatgpt-openai-microsoft-meta-midjourney-other-ai-cos/ (references to Damle as attorney for various AI defendants).
[98] Damle, *supra* note 2, at 7.
[99] Lemley & Casey, *supra* note 7*,* at 772; *see also* R Street, *supra* note 75 (asserting that AI systems are designed to "learn facts about the world, ideas and visual concepts" rather than to exploit expressive

Reassuring depictions of the AI training process such as these amount to sleights of hand that distract from the reality that expressive content is being stored in the model. As explained above, training materials do not disappear as the model is built; rather, each work is algorithmically ingested, piece by piece—or token by token—into the model.[100] Nor is there any practice of separating copyrightable from uncopyrightable elements during the training process. The tokens themselves are encoded—not just "statistical data" or "information" about them.[101] Of course, this is only logical; the whole point of the training exercise is to capture and map the expressive content of each work for use in the generative process.

In this regard, it is worth pointing out that, ultimately, information about a work *is* the work. For example, if I were to provide you with a table listing all the words in a book, which identified the first word as "It," the second as "is," the third as "a," the fourth as "truth," the fifth as "universally," the sixth as "acknowledged," and so on, all the way to the end of the book—and instructed you to transcribe the words accordingly to the order specified in the table—after a (very long) time you would have *Pride and Prejudice.*[102] Or I could provide you with the color and location of every pixel in a family photo and ask you to recreate it from those data.[103] In either case, you would be reproducing the original work using "information about the work."

In ordinary usage, "information about a work" means data derived from a work *that exist separately from the work*, such as the location of a particular word within a text, the size of an image file, the length of a song, and so forth.[104] It does not mean an encoded version of the work that can be tapped to produce a representation of the work or generate additional works. As more fully discussed below, the distinction between expressive versus nonexpressive use of protected works was critical in the *Google Books* case (among others), where the court's determination of fair use turned on its conclusion that displaying brief "snippets" of text indicating where particular words appeared in a book was not a substitute for the book itself.[105] The *Google*

content). *But see* Lemley & Casey, *supra* note 7, at 777 (contradictorily observing that "[s]ome ML systems will be interested in the expressive components of the work as an integral part of their training. That is, the goal will be to teach the system using the creative aspects of the work that copyright values, not just using the facts or the semantic connections the law is not supposed to protect.").

[100] *See supra* notes 42-54 and accompanying text (describing encoding process).

[101] *See supra* notes 44-46 and accompanying text (discussing role of tokens).

[102] Jane Austen, *Pride and Prejudice* 1 (Bantam Classic ed., Random House, Inc. 2003) (1813) ("It is a truth universally acknowledged that a single man in possession of a good fortune must be in want of a wife.")

[103] Sobel cites a similar example to demonstrate that conventional image files consist of mathematical representations: "An image file might instruct a computer, 'display a 50x50 pixel grid of alternating rows of white and red pixels.' That's a mathematical representation of a white-and-red-striped square." Sobel, *Elements*, *supra* note 79, at 24.

[104] *See Google Books*, 804 F.3d at 217, 225 (describing purpose of Google's copying as "to make available significant information *about those books*," including "whether[] and how often they use specified words or terms") (emphasis in original).

[105] *Id.* at 224-25.

Books example stands in marked contrast to the exploitation of copyrighted works by AI systems and users of those systems, who are utilizing stored expressive content to produce output.

The encoding of text, images and music into machine-readable formats is, of course, nothing new. Copyright law protects works stored as code and in other digital formats.[106] As explained above, to convert a work from one format to another is to make a reproduction of the work (and not a basis for a claim of fair use).[107] No one would argue, for example, that saving a Word document as a PDF, or an analog photo to a JPG file, alters the protectability of the work in its original format. Indeed, as recognized by courts and the Copyright Office, the computer code embodying an aesthetic work and a rendering of the work generated by that code can properly be considered one and the same work.[108]

Copyright law also protects works that have been digitally stored in segments or pieces. In the file-sharing case *Columbia Picture Industries, Inc. v. Fung*,[109] for instance, the Ninth Circuit did not hesitate to hold that the unauthorized uploading and downloading of motion pictures using BitTorrent technology—which breaks works into fragments for rapid transmission among simultaneous users—was infringing.[110] Cloud storage services rely on analogous technology, dividing files into smaller "blocks" for hosting in multiple locations, from which locations they are more efficiently retrieved (and reconstructed) for users.[111] It would seem foolish to suggest that a work stored in the cloud loses copyright protection.

Of course, in the context of generative AI, a training work is encoded into an AI model along with millions of other works, all of which contribute to the generative capacity of the model.

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[106] "Copyright protection subsists, in accordance with this title, in original works of authorship fixed in any tangible medium of expression, now known or later developed, from which they can be perceived, reproduced or otherwise communicated, either directly or with the aid of a machine or device." 17 U.S.C. § 102(a); *see also Green v. U.S. Dep't of Justice*, No. 23-5159, 2024 U.S. App. LEXIS 19261, at *7 (D.C. Cir. Aug. 2, 2024) (Congress enacted the Digital Millennium Copyright Act to protect "'the movies, music, software, and literary works that are the fruit of American creative genius'" from digital piracy (quoting S. Rep. No. 105-190, at 8 (1998))).

[107] *See supra* notes 93-96 and accompanying text.

[108] *See* U.S. Copyright Office, *Compendium of U.S. Copyright Office Practices* § 721.10(A) (computer program and the screen displays generated by that program "are considered the same work, because the program code contains fixed expression that produces the screen displays"); *see also Williams Elecs., Inc. v. Artic Int'l, Inc.*, 685 F.2d 870, 874 (3d Cir. 1982) (changing videogame display protectable because fixed in memory device by which it was generated) (citing *Stern Elecs., Inc. v. Kaufman*, 669 F.2d 852, 855-56 (2d Cir. 1982) and *Midway Mfg. Co. v. Artic Int'l, Inc.*, 547 F. Supp. 999, 1006-07 (N.D. Ill. 1982), for same principle). Of course, computer code that embodies and renders a particular artistic work is distinct from software (such as Microsoft Word) that is used as a tool to *create* such a work; the two are separately protectable works.

[109] 710 F.3d 1020 (9th Cir. 2013).

[110] *Id.* at 1026, 1034.

[111] *See* Cloudflare, *What is block storage?* (last visited June 29, 2024), https://www.cloudflare.com/learning/cloud/what-is-block-storage/.

Due to limitations of generative AI technology, it seems impossible to extract a particular work from a model once it has been ingested, at least at this point in time.[112] But the fact that a particular work co-exists with myriad others—and that the encoding process does not permit *post hoc* removal of particular works—does not excuse the infringement of that one work. The illicit copying of a single work is not excused by illicit copying of others.[113] AI companies seek to minimize the significance of copying of individual works by asserting, for example, in the LLM context, that it is "the *volume* of text used, more than any particular *selection* of text, that really matters."[114] But copying more does not mean you are copying less. In the words of the Supreme Court, "a taking may not be excused merely because it is insubstantial with respect to the *infringing work*."[115]

Thus understood, and as suggested by Professor Lee and her co-authors, far from being a work created independently of the works it trained on through some sort of detached "learning" process, an AI model is appropriately considered a derivative or compilation of the works it embodies.[116]

3. ***Regeneration of training works in output***

AI fair use advocates do not deny that AI-generated output that is substantially similar to copyrighted works on which the system was trained is a problem. Nor do they seem to be asserting that such copies—which could serve as substitutes for the originals—fall within the ambit of fair use. Instead, the fair use camp maintains that the propensity of AI models to reproduce memorized copies of training materials in output should be viewed as an "aberration" or "quirk"[117]—that is, as inconsequential collateral damage that AI companies are trying to ameliorate and which should not be held against them.[118]

As discussed above, the generation of infringing copies does not seem especially rare. In fact, absent the development of truly effective filtering systems, it would appear to be a possibility

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[112] Stephen Pastis, *A.I.'s unlearning problem: Researchers say it's virtually impossible to make an A.I. model 'forget' the things it learns from private user data*, Fortune (Aug. 30, 2023), https://fortune.com/europe/2023/08/30/researchers-impossible-remove-private-user-data-delete-trained-ai-models/ ("'If a machine learning-based system has been trained on data, the only way to retroactively remove a portion of that data is by re-training the algorithms from scratch.'") (quoting New York University computer scientist Anasse Bari)).

[113] *See, e.g.*, *Sony Music Entm't v. Cox Commc'ns, Inc.*, 93 F.4th 222, 236-37 (2024) (upholding infringement claims in case involving mass infringement of numerous plaintiffs' copyrighted musical works and sound recordings).

[114] *Tremblay* MTD, *supra* note 2, at 2 (emphasis in original); *see also* Damle, *supra* note 2, at 13 (arguing that AI companies must operate without licenses due to the need to train on "virtually the entire internet").

[115] *Harper & Row, Publishers, Inc. v. Nation Enters.*, 471 U.S. at 539, 565 (1985) ("*Harper & Row*") (emphasis in original) (copied material, though comprising a small part of defendant's work, was infringing).

[116] *See infra* notes 182-84 and accompanying text.

[117] *See, e.g.*, Damle, *supra* note 2, at 8.

[118] *See supra* note 75 and accompanying text (AI companies and advocates characterizing generation of infringing output as rare occurrence).

with respect to any work in the training set should the right prompt (or other method of extraction) be employed.[119] In any event, regardless of how often "regurgitation" occurs, or the particular prompts that cause it to manifest, even AI companies seem to acknowledge that such output is appropriately considered infringing.

4. *RAG activities*

Last but not least, as explained above, RAG-enabled AI systems ingest content from websites and other sources to feed into user queries in order to augment the substance and relevance of system output. It is thus apparent that RAG technology is exploiting third-party content for its expressive value.[120] At the same time, RAG has only recently emerged as a significant mode of unauthorized exploitation. We have not yet heard much from AI companies regarding a legal justification for the ongoing unlicensed copying required to enable RAG functionality.

III. Fair Use Analysis

Section 107 of the Copyright Act sets forth the factors courts are to consider in evaluating a claim of fair use.[121] A review of those factors and relevant decisional law confirms that there is no established principle of fair use that sanctions the mass reproduction and exploitation of copyrighted works to develop, build and operate AI systems for profit.

A. The cases relied on by AI companies

Pointing to a handful of technology-driven fair use cases, AI companies and their advocates claim that large-scale reproduction of copyrighted works to develop and populate AI systems constitutes a fair use of those works.[122] But *Google Books*, *HathiTrust*, *Sega* and other key

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[119] *See supra* notes 69-76 and accompanying text (discussing memorization and challenges of filtering).
[120] *See supra* notes 82-85 and accompanying text.
[121] Section 107 provides in pertinent part as follows:

> [T]he fair use of a copyrighted work, including such use by reproduction in copies or phonorecords or by any other means specified by that section, for purposes such as criticism, comment, news reporting, teaching (including multiple copies for classroom use), scholarship, or research, is not an infringement of copyright. In determining whether the use made of a work in any particular case is a fair use the factors to be considered shall include—
> (1) the purpose and character of the use, including whether such use is of a commercial nature or is for nonprofit educational purposes;
> (2) the nature of the copyrighted work;
> (3) the amount and substantiality of the portion used in relation to the copyrighted work as a whole; and
> (4) the effect of the use upon the potential market for or value of the copyrighted work.

17 U.S.C. § 107.
[122] *See supra* notes 2-7 and accompanying text (summarizing fair use claims by AI companies and their representatives).

precedents relied upon by AI companies to defend their unlicensed copying[123]—mainly *Kelly v. Arriba Soft Corp.*,[124] *Perfect 10, Inc. v. Amazon.com, Inc.*,[125] *A.V. v. iParadigms, LLC* ("*iParadigms*"),[126] *Sony Computer Entertainment, Inc. v. Connectix Corp.* ("*Sony Computer*")[127] and *Google, LLC v. Oracle America, Inc.* ("*Oracle*")[128]—are all in a different category with respect to fair use. That is because these cases were concerned with functional rather than expressive uses of copied works. The copying challenged in each was to enable a technical capability such as search functionality or software interoperability. By contrast, copying by AI companies serves to enable exploitation of protected expression.

1. ***Earlier cases address different conduct***

AI companies seem to place their greatest faith in the mass book copying cases, *Google Books* and *HathiTrust*. But to claim that these Second Circuit decisions legitimize the sorts of copying engaged in by AI systems is to take their carefully limited holdings a bridge unquestionably too far.

There was no general declaration in either *Google Books* or *HathiTrust* that mass reproduction of copyrighted works to construct a product predicated upon on large-scale copying has any presumptive claim to fair use. To the contrary, the *Google Books* panel was careful to cabin its holding to the particular circumstances before it,[129] including the fact that Google's search functionality returned only snippets of text that did not permit meaningful consumption of expressive content.[130] Although Google made full-text copies of the books, it was not seeking to capitalize on, or allow users to exploit, the aesthetic value of those works. Even so, the court considered Google's copying to "test the boundaries of fair use."[131] Indeed, the court pointedly observed that had Google permitted users greater access to "the expressive content of the

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[123] *See* Damle, *supra* note 2, at 5-8 (characterizing *Google Books, HathiTrust* and *Sega* as "foundational precedents" that establish the use of copyrighted works to develop AI systems as "quintessentially fair"), 5-6 nn.10-23 (citing additional technological copying cases); Sag, *Copyright Safety*, *supra* note 2, at 304-09 (asserting legitimacy of AI copying under *Google Books* and *HathiTrust*); Sag, *Fairness*, *supra* note 24, at 1903-06, 1913-14 (interpreting technological fair use cases to support claim that AI copying is fair use).
[124] 336 F.3d 811 (9th Cir. 2003).
[125] 508 F.3d 1146 (9th Cir. 2006).
[126] 562 F.3d 630 (4th Cir. 2009).
[127] 203 F.3d 596 (9th Cir. 2000).
[128] 593 U.S. 1 (2021).
[129] *See Google Books*, 804 F.3d at 207, 222, 224-25, 229 (qualifying its holding with terminology such as "at least under present conditions," "in these circumstances," "at this time," "at least as presently structured," "as … presently constructed," "at least as … presently designed," "[o]n the present record," etc.); *see also* Matthew Sag, *The New Legal Landscape for Text Mining and Machine Learning*, 66 J. Copyright Soc'y of the U.S.A. 291, 294 (2019), file:///C:/Users/user/Downloads/SSRN-id3331606.pdf (noting that *Google Books* and *HathiTrust* "were a product of the particular factual circumstances").
[130] *See supra* note 39 and accompanying text (describing snippet functionality).
[131] *Google Books*, 804 F.3d at 206.

original," such exploitation "would most likely constitute copyright infringement if not licensed by the rights holders."[132]

The determination of fair use in *HathiTrust* was similarly confined to the facts of that case. There, the search results were even more limited, listing only the page numbers on which, and number of times, a specific term appeared in the relevant text.[133]

In both *Google Books* and *HathiTrust*, the court established a dividing line between uses that were functional and nonsubstitutional in nature, and uses that were not. Despite its superficial similarity to the mass copying in *Google Books* and *HathiTrust*, then, AI copying cannot be squared with the fair use finding in either of these cases, the holdings of which were careful to preserve copyright owners' legitimate interest in the expressive content of their works.

In *Kelly*, an earlier technological copying case, a professional photographer challenged a search engine's reproduction and indexing of images taken from his own and licensee websites.[134] In responding to user queries, the engine returned small, low-quality reproductions that users could click on to access linked, full-sized versions of the works.[135] The court held that the defendant's copying of photos to provide a search and indexing service was a transformative fair use because the low-quality thumbnails "serve[d] a different function" than the originals—namely, "improving access to information on the internet."[136] The court pointedly distinguished this purpose from copying to capitalize on "artistic expression."[137] It found the search engine's use to be nonsubstitutional because, unlike Kelly's photos, the search and indexing function was "unrelated to any aesthetic purpose."[138]

Like *Kelly*, *Perfect 10* involved unlicensed copying and indexing of online images, in this case by Google, which displayed thumbnails of Perfect 10's copyrighted photos to users of its search technology.[139] Users similarly could access the full-sized images via a link to the originating website.[140] Invoking *Kelly*, the Ninth Circuit once again held that a search engine's copying of images for thumbnail display was a transformative fair use because the images were not being used for their intrinsic purpose but to create "an electronic reference tool."[141]

In each of the above cases, the fair use determination turned on the fact that the defendant was not exploiting expressive content for its intrinsic aesthetic value. So, too, in *iParadigms.*

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[132] *Id.* at 226.
[133] *HathiTrust*, 755 F.3d at 91. The *HathiTrust* court pointedly observed that its determination was made without foreclosing future claims "based on a different record." *Id.* at 101.
[134] *Kelly*, 336 F.3d at 815-16.
[135] *Id*.
[136] *Id.* 818-19.
[137] *Id.* at 819.
[138] *Id.* at 818-19.
[139] *Perfect 10*, 508 F.3d at 1155-56.
[140] *Id.*
[141] *Id.* at 1164-65,

iParadigms considered student authors' challenge to iParadigms' "Turnitin" plagiarism detection service, which digitally compared school papers against an archive of previously submitted papers to identify instances of copying.[142] The student plaintiffs alleged that the archiving of their papers in the Turnitin database violated their copyrights.[143] Here again, the court focused on the fact that the defendant's use of the copied content "had an entirely different function and purpose than the original works," emphasizing that the use was "unrelated to any creative component" of the student works.[144] The fact that iParadigms was not seeking to exploit student essay content as such readily distinguishes the copying at issue in *iParadigms* from that engaged in by generative AI systems.

Sega, Sony Computer and *Oracle* are even farther afield from the types of copying engaged in by AI companies. In each of these cases, the copying was undertaken with respect to a specific work to facilitate interoperability—clearly not the objective of AI copying.

In *Sega*, for instance, defendant Accolade copied Sega's videogame code in order to "reverse engineer" it to identify the functional requirements that would allow Accolade to develop its own videogames that would play on Sega consoles.[145] As a result of this process, Accolade identified a small segment of code in Sega's program that served to limit the game's operability to Sega consoles.[146] Although Accolade had engaged in "wholesale" copying of the videogame program to identify the digital lock, it included only the brief segment of functional code (for which Sega did not pursue a separate claim of infringement) in its own product.[147] The Ninth Circuit held Accolade's "intermediate" copying of Sega's program to be a fair use because it served a "legitimate, essentially non-exploitative purpose"—that is, to discover the functional aspects of Sega's code in order to produce compatible works.[148] Observing that intermediate copying was "the only means" by which Sega could access unprotected elements of the console code, the court expressed concern that a finding of infringement would grant Sega a "*de facto* monopoly" over functional aspects of the code.[149] In concluding that the reverse engineering was fair use, the court also emphasized that Accolade's works did not incorporate any of Sega's creative expression.[150] The intermediate copying in *Sega* cannot be equated with copying by AI systems, the very different purpose of which is to capture and use expressive content.

The Ninth Circuit adhered to the *Sega* precedent in *Sony Computer*, another reverse engineering case in which the court held that the defendant's intermediate copying of plaintiff Sony's console code to create a "virtual" gaming station that would allow users to play Sony games on a

[142] *iParadigms*, 562 F.3d at 634.
[143] *Id.* at 635.
[144] *Id.* at 639, 641-42.
[145] *Sega*, 977 F.2d at 1514-15, 1522.
[146] *Id.* at 1516.
[147] *Id.*
[148] *Id.* at 1522-23.
[149] *Id.* at 1523-27.
[150] *Id.* at 1570 ("[T]here is no evidence in the record that Accolade sought to avoid performing its own creative work").

personal computer was a fair use.[151] Once again, the holding was premised on the court's determination that the code at issue "contain[ed] unprotected functional elements" that could not be accessed or studied without copying.[152]

Such intermediate copying—an interim step in the software development process to ascertain functional requirements for a technical purpose—is not comparable to the copying that occurs in the context of generative AI. As its name would suggest, intermediate copying does not involve ongoing engagement with or exploitation of protected content. The defendants in *Sega* and *Sony Computer* were not seeking to replicate or profit from the plaintiffs' artistic works but instead to produce independently created, compatible products.[153] An AI model, by contrast, is designed permanently to exploit copied creative expression.

Finally, in *Oracle*, the Supreme Court concluded that Google's copying of Java "declaring" code—a widely used system to call up computer functions—so coders could more easily write programs for Android phones was a fair use.[154] In so holding, the Court emphasized the functional nature of the code appropriated by Google, the copyrightability of which it found questionable.[155] Google's copying targeted a widely used, utilitarian aspect of computer code that the Court viewed as far from the "core of copyright's protection."[156] The Court was persuaded that Google did not copy the lines of Java code "because of their creativity, their beauty, or even (in a sense) because of their purpose," but so Java-trained programmers writing software for Android phones could rely on commands that "they [were] already familiar with to call up particular tasks."[157] In other words, the Court was focused on the interoperability of human programmers.[158]

In short, none of the fair use precedents on which AI companies purport to rely addressed a product designed to copy and exploit third-party expressive content to derive new content,

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[151] *Sony Computer*, 203 F.3d at 601, 608.
[152] *Id.* at 603-04.
[153] *See Sega* at 1518 (intermediate copying of object code fair use where it was only means of access to elements not protected by copyright); *see also Sony Computer* at 598 (resulting product did not contain any copyright-protected material).
[154] *Oracle*, 593 U.S. at 11-15.
[155] *See id.* at 29 ("[T]he declaring code is, if copyrightable at all, further than are most computer programs ... from the core of copyright."). In a nuanced analysis of the majority opinion, Professor Jane Ginsburg suggests that "[i]n effect, the fair use determination achieved the same result as ruling the [copied code] uncopyrightable, but attained that objective through the back end of a copyright exception rather than the front end of applying the idea/expression distinction …."). Jane C. Ginsburg, *Fair Use in the US Redux*, Singapore J. Legal Stud. 3 (Mar. 2024), https://law.nus.edu.sg/sjls/wp-content/uploads/sites/14/2024/05/firstview-march24-JaneGinsburg.pdf.
[156] *Oracle*, 593 U.S. at 29.
[157] *Id.* at 13-14.
[158] *See id.* (Google copied Oracle's code "because programmers had already learned to work with the [Java] system, and it would have been difficult, perhaps prohibitively so, to attract programmers to build its Android smartphone system without them.").

including potentially competing and infringing content. None involved copying and use of expressive content for its intrinsic value.

This critical distinction between expressive and nonexpressive use of copyrighted works in a technological context is well illustrated by the Second Circuit's post-*Google Books* decision in *Fox News Network v. TVEyes, Inc.*[159] In that case, the court considered an unlicensed service that recorded "essentially all television broadcasts" and the closed-caption text that accompanied them in order to create a text-searchable database of the copied video. By typing in a search term, users would receive a list of news clips in which the term appeared, which references could then be clicked on and played.[160]

In defending its copying, TVEyes relied primarily on *Google Books.*[161] But the court firmly distinguished that precedent to hold TVEyes' systematic exploitation of Fox's news content to be an infringing, rather than fair, use.[162] Significantly, the court's rejection of fair use was predicated on its core finding that the use of Fox's content was "both 'extensive' and inclusive of all that is 'important' from the copyrighted work."[163] Unlike in *Google Books*, TVEyes had built a business based on the appropriation of expressive content:

> The success of the TVEyes business model demonstrates that deep-pocketed consumers are willing to pay well for a service that allows them to search for and view selected television clips, and that this market is worth millions of dollars in the aggregate. Consequently, there is a plausibly exploitable market for such access to television content, and it is proper to consider whether TVEyes displaces potential Fox revenues ….[164]

Taking note of the Second Circuit's cautionary language in *Google Books* that, notwithstanding the guardrails implemented by Google to prevent expressive use of the copied books, Google's conduct "'test[ed] the boundaries of fair use,'" the *TVEyes* court concluded that defendant *TVEyes* had "exceeded those bounds."[165]

2. *There is no presumption that a use is fair if the end product is noninfringing*

A favored argument of those advocating on behalf of AI companies is that the appropriation of copyrighted works to train AI systems is a fair use of those works so long as the resulting product does not infringe copied expression. Damle, for instance, states in his congressional testimony

[159] 883 F.3d 169 (2d Cir. 2018).
[160] *Id.* at 175.
[161] *Id.* at 177.
[162] *Id.* at 174-75, 179-81.
[163] *Id.* at 179.
[164] *Id.* at 180.
[165] *Id.* at 174 (quoting *Google Books*, 804 F.3d at 206).

that "[a]n unbroken line of cases establishes that the use of a copyrighted work to create a non-infringing final product is quintessential fair use."[166] Sag puts it a little differently, avowing that U.S. fair use cases "have consistently held that technical acts of copying which do not communicate an author's original expression to a new audience are fair use."[167] The theory seems to be unless an outsider can see the infringement, there is no infringement.

In advocating for this approach, Damle is not entirely clear as to whether he means the resulting model is noninfringing, its output is noninfringing, or both. For the reasons discussed above, however, none of these is a valid proposition.[168] It is not reasonable to assume that the AI model itself is not infringing or that it will not communicate authors' original expression, in all or in part, to users when generating output.[169]

In any event, no court has ever articulated a canon of fair use such as that posited by Damle and Sag. Indeed, the Ninth Circuit's *Sega* opinion, on which both rely,[170] expressly *rejected* any such standard. Addressing a similar argument by the defendant—that its intermediate copying had to be fair because its final product was noninfringing—the court was clear that "intermediate copying of computer object code may infringe the exclusive rights granted to the copyright owner in section 106 of the Copyright Act *regardless of whether the end product of the copying infringes those rights*."[171] This principle was reaffirmed by the Ninth Circuit nearly twenty years later in *Sony Computer*,[172] another case often invoked to justify AI copying.[173] In short, there is no "unbroken line" line of fair use precedent that immunizes AI copying as noninfringing if the generated output is considered noninfringing.

B. AI copying under the fair use factors

1. *The purpose and character of AI copying*

The first fair use factor of section 107 considers the purpose and character of the challenged use, including—under judge-made law—whether the use is "transformative," and whether it is for a commercial purpose.[174] The advocates of AI fair use do not claim that the training of for-profit

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[166] Damle, *supra* note 2, at 5, 7.
[167] Sag, *Fairness*, *supra* note 24, at 1903. Sag relatedly contends that AI copying is a "nonexpressive" use of the copied works. *See id.* at 1914; Sag, *Copyright Safety*, *supra* note 2, at 307-09.
[168] *See supra* notes 97-119 and accompanying text (discussing encoding of training works in AI model and their regeneration in output).
[169] *See supra* notes 60-68 and accompanying text (discussing replication of training works).
[170] *See* Damle, *supra* note 2, at 6; Sag, *Fairness*, *supra* note 24, at 1904.
[171] *Sega*, 977 F.2d at 1517-19 (emphasis added). The court instead grounded its fair use finding on the fact that the copying was undertaken solely to identify functional elements rather than to exploit Sega's creative expression. *Id.* at 1522-23.
[172] *See Sony Computer*, 203 F.3d at 602-03 ("In *Sega*, we recognized that intermediate copying could constitute infringement even when the end product did not itself contain copyrighted material.")
[173] *See, e.g.*, Damle, *supra* note 2, at 6 n.21; Sag, *Fairness*, *supra* note 24, at 1903 n.100.
[174] *See Warhol*, 598 U.S. at 527-29; *Campbell*, 510 U.S. at 578-79.

AI models falls within any of the favored purposes listed in section 107—criticism, comment, news reporting, teaching, scholarship, or research.[175] Instead, they look to the technological fair use precedents discussed above to argue that the use is transformative. As shown, however, those cases were careful to distinguish copying and exploitation of expressive content for its intrinsic value from the technological copying they found to be fair. These cases thus point in the opposite direction of fair use with respect to copying by AI entities, as do the purpose and character of such copying.[176]

a. *AI copying is not transformative*

For purposes of fair use, a transformative use is one that "'adds something new, with a further purpose or different character'" and does not supplant the original.[177] In *Google Books*, for example, the Second Circuit held that the digital scanning of books in order to provide a search function was transformative because it "augments public knowledge by making available information about Plaintiffs' books without providing the public with a substantial substitute for matter protected by the Plaintiffs' copyright interests in the original works or derivatives of them."[178] The *HathiTrust* court reached a similar conclusion, holding that the copying of books to create a searchable database was a transformative use that "add[ed] … something new with a different purpose and a different character."[179] In *Sega*, a competing videogame manufacturer disassembled Sega's console code, not to copy Sega's creative expression, but to develop its own original, compatible games.[180] The "transformativeness" of the use in each of these cases turned on the fact that aesthetic content was not being exploited for its own sake—either by defendants themselves or users of their products.

Copying of protected works by generative AI systems has no similar claim to transformativeness. Works are copied in their entirety and mechanically encoded in the AI model without offering any search functionality or other utility to users, let alone criticism or commentary. They are copied to capture their expressive value.

Fair use proponents contend that the training process merely records "unprotected facts" about the training works but, as shown above, that is not the case.[181] In fact, the AI model maps and stores the expressive content of each work so it can be tapped to enable the model's generative capabilities. That the works are parsed into small segments, or "tokens," and mathematically mapped into vectors, does not negate the appropriation of expressive content.

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[175] *See* 17 U.S.C. § 107.
[176] In addition to being relevant to claims of direct infringement, the purpose and design of AI models present significant issues of secondary copyright liability arising from user activities, a rich subject in itself that is beyond the scope of this paper.
[177] *Warhol*, 598 U.S. at 528 (quoting *Campbell*, 510 U.S. at 579).
[178] *Google Books*, 804 F.3d at 207.
[179] *HathiTrust*, 755 F.3d at 97-98.
[180] *Sega*, 977 F,2d at 1522-23.
[181] *See supra* notes 100-05 and accompanying text.

A process that mechanically converts works so they can be exploited in a more convenient format does not qualify as a transformative purpose. To be sure, copyrighted works are "transformed" when they are encoded in an AI model, but not in the senses of fair use "transformativeness"—rather, in the sense of creating a derivative. Under the Copyright Act, the owner of a protected work has the exclusive right to prepare, or authorize the preparation of, a derivative of that work,[182] defined in the Act as "a work based upon one or more preexisting works," including "any … form in which a work may be recast, transformed or adapted."[183] An AI model, created by reproducing and encoding—or "recasting"—copyrighted works into the model, falls within this definition. An AI model may also be considered a compilation, that is, "a work formed by the collection and assembling of preexisting materials or of data that are selected, coordinated, or arranged in such a way that the resulting work as a whole constitutes an original work of authorship."[184] Whether a derivative work or a compilation, however, the model makes unauthorized use of copyrighted works.

Fair use advocates may assert that, in assessing transformativeness, that the purpose of the copying to be considered is not that of building and operating AI models—its immediate objective—but the ultimate goal of allowing users to generate new content from the works encoded in the models. This alternative argument for transformativeness falls short, for several reasons.

To begin with, the generation of new content by an AI model based on a user prompt is inextricably bound up with the expressive exploitation of the copied works. It is not a content-neutral function akin to search capability or interoperability. As explored above, AI machines do not "think" independently; their production of new content is a function of and limited by the creative content they have ingested.[185] Accordingly, the use of copyrighted works to facilitate AI generation does not align with the reasoning of *Google Books*, *HathiTrust* or other technological cases in which the copying was found to be transformative because it did not exploit expressive content.

Next, as discussed above, AI machines not infrequently generate infringing copies and derivatives of the works on which they are trained.[186] While this may not be a desired outcome, it appears to be an unavoidable feature of AI systems, at least at present.[187] To state the obvious, AI companies cannot claim transformativeness based on output that merely reproduces all or part of a training work or works, even if it is done in a complicated way. An AI model's generation of text or an image that is a copy of a training work and could serve as substitute for the original is

[182] 17 U.S.C. § 106 ("[T]he owner of a copyright … has the exclusive right[] to … prepare derivative works based upon the copyrighted work.").
[183] 17 U.S.C. § 101 (definition of "derivative work").
[184] 17 U.S.C. § 101 (definition of "compilation").
[185] *See supra* notes 18-31 and accompanying text.
[186] *See supra* notes 60-68 and accompanying text.
[187] *See supra* notes 71-73 and accompanying text.

presumptively nontransformative.[188] Nor is there anything inherently transformative about combining elements of one work with those of another work or works, which implicates the copyright owners' derivative work rights.[189] As Professor Jane Ginsburg observes, "AI outputs may incorporate the source works' expression in a new production; but that output generally will not comment, criticize, shed light on or otherwise be *about* the copied expression."[190]

Notably, in *Andy Warhol Foundation for the Visual Arts, Inc. v. Goldsmith*,[191] the Supreme Court warned against "an overbroad concept of transformative use" that encroaches upon copyright owners' derivative work rights, explaining that an interpretation of transformativeness "that includes any further purpose, or any different character" could "swallow" the copyright owner's exclusive right to create derivative works.[192] To this end, the Court criticized overzealous application of "transformativeness" to encompass any work that "adds some new expression, meaning, or message."[193] Drawing on its earlier explication of fair use in *Campbell*,[194] the Court emphasized that the secondary user must have an independent justification for use of the work in question. That a copied work may be useful to convey a new meaning or message is not justification enough.[195]

AI-generated content that is not a recognizable copy or derivative of a training work or works—that is, the type of content AI companies claim to be the intended output of their systems—by definition does not comment or shed light on any particular work. It is therefore difficult to see how AI entities can stake a claim to transformative use of training works based on output that does not convey commentary or criticism with respect to those works. Nor, as noted, does AI output facilitate a utility-expanding function such as searchability or indexing.

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[188] *See Warhol*, 598 U.S. at 532-33 (first fair use factor likely to weigh against fair use where "an original work and a secondary use share the same or highly similar purposes, and the secondary use is of a commercial nature").
[189] *See, e.g.*, *Warhol*, 598 U.S. at 537, 550-51 (2023) (unlicensed commercial use of plaintiff's photograph, as incorporated into an Andy Warhol silkscreen derivative, was nontransformative and therefore infringing); *Dr. Seuss Enters., L.P. v. ComicMix LLC*, 983 F.3d 443, 451-55 (9th Cir. 2020) ("*Seuss*") (ComicMix's unlicensed book consisting of a "mashup" of Dr. Seuss and Star Trek characters that mimicked Dr. Seuss illustrations was a nontransformative use of Seuss's works).
[190] Ginsburg, *supra* note 155*,* at 29.
[191] 598 U.S. 508 (2023).
[192] *Warhol*, 598 U.S. at 529, 541. In keeping with this instruction, the Court determined that a magazine's commercial use of a silkscreen image created by Andy Warhol from plaintiff Goldsmith's photographic portrait of Prince was not transformative because it served as a substitute for Goldsmith's original photo. *Id*. at 523-24.
[193] *Id.* at 541.
[194] In *Campbell*, the Court considered 2 Live Crew's use of Roy Orbison's classic song "Pretty Woman" in a rap parody, finding the use transformative because it was necessary to copy portions of Orbison's work in order to mock it. *See Campbell*, 510 U.S. at 579-83.
[195] *See Warhol*, 598 U.S. at 530-33 ("If an original work and a secondary use share the same or highly similar purposes, and the secondary use is of a commercial nature, the first factor is likely to weigh against fair use, absent some other justification for copying."), 546-48 ("Copying might [be] helpful to convey a new meaning or message. It often is. But that does not suffice under first factor.").

This leaves us with the bare claim that AI copying should be considered transformative because it enables the generative capabilities of AI models. This broad contention is untethered to the use of any particular work or works, but instead boils down to an assertion that mass appropriation of protected works is justified because extensive copying is necessary to build and operate such systems. In effect, then, it amounts to a policy argument that the rights of copyright owners must yield to the presumed social benefits of generative AI technology.

Courts sometimes consider the public benefit of a challenged use in evaluating the question of transformativeness. In *Google Books*, for instance, the court determined that "Google's making of a digital copy to provide a search function is a transformative use, which augments public knowledge by making available information *about* Plaintiffs' books without providing the public with a substantial substitute."[196] In *Sega*, the court was concerned that a failure to permit reverse engineering would "confer[] on the copyright owner a *de facto* monopoly" over unprotectable ideas and functional concepts in its computer code that others could build upon.[197] In *Oracle*, the Court reasoned that utilizing Java declaring code would enable Java-trained programmers to "expand the use and usefulness" of Android phones.[198]

AI copying does not fit within these paradigmatic examples of technologically driven copying. To begin with, none of these cases held the extraction of creative value from protected works to be a transformative use. Moreover, unlike in the case of generative AI, in each of these cases, the beneficial purpose could not have been achieved without copying the specific work or works at issue. That is, there was a clear nexus between the appropriated works and the claimed social benefit. In *Google Books*, for example, Google could not have provided a search function that would locate a term in a book without copying the book in question. In *Sega*, the court determined that copying Sega's console code was necessary to access the functional aspects of the code in order to create independent games. The same was true in *Oracle*; the only way to harness the collective knowledge of Java programmers was to emulate the Java command structure. The same cannot be said of generative AI systems, which copy millions of works in an indiscriminate fashion. The generalized nature of AI copying is inconsistent with claims that the copying serves a transformative purpose vis-à-vis the copied works.[199]

The claim that AI companies' mass copying is transformative seems largely premised on the view that generative AI is remarkable technology with the power to enhance creativity and improve society.[200] But a generalized assertion of public good such as this does not justify the

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[196] *Google Books*, 804 F.3d at 207 (emphasis in original).
[197] *Sega*, 977 F.3d at 1573-74.
[198] *Oracle*, 593 U.S. at 30.
[199] *Cf. Seuss*, 983 F.3d at 454 (finding no transformative use where defendant ComicMix had no particular need to use Seuss's material for its story).
[200] *See, e.g.*, Damle, *supra* note 2, at 1 ("The AI tools of the present and near future will impact almost every aspect of the human experience …. They will transform the way humans learn and work. They will enable anyone to more fully unlock their creative potential. In short, AI has the potential to transform our economy and improve our society as a whole."); Wayne Brough & Ahmad Nazeri, *Regulatory*

use of particular copyrighted works or qualify as a transformative purpose.[201] If it did, fair use could be claimed with respect to any beneficial technological advance that sought to capitalize on copyrighted works for free.[202]

Finally, a claim of transformative purpose is especially weak where, as here, the unauthorized copying is supplanting the licensed use of copyrighted works for the same purpose—namely, to train and operate AI systems—as discussed below.[203] As the Supreme Court emphasized in *Warhol*, where use of an original work and a secondary use "share the same or highly similar purposes, and the secondary use is of a commercial nature, the first factor is likely to weigh against fair use, absent some other justification for copying."[204] AI companies have not articulated any justification for their mass unauthorized copying apart from generalized assertions that it is necessary to build their systems and that licensing would be too difficult.[205] Neither of these constitutes a transformative purpose under copyright law. Moreover, as shown below, the second claim in particular is contrary to the facts.

b. ***Commercial purpose of copying***

Leading AI developers, including OpenAI, Stability AI and Anthropic, not to mention the tech giants Google, Meta and Microsoft, indisputably engage in for-profit activities, including the sale of AI products to the public.[206] The use of copyrighted works to develop and operate a

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Comments Before the U.S. Copyright Office Library of Congress In the Matter of Artificial Intelligence and Copyright, R Street (Oct. 30, 2023), https://www.rstreet.org/outreach/regulatory-comments-before-the-u-s-copyright-office-library-of-congress-in-the-matter-of-artificial-intelligence-and-copyright/ ("AI promises to bolster the American economy, amplify the capabilities of creatives and catalyze advancements in science and the arts.")

[201] On the question of public benefit, it is worth noting that although generative AI may have the potential for positive impact in various areas of human endeavor, it comes with significant social concerns, among them its ability to generate false information; impersonate individuals (including by producing "deepfakes"); and amplify racism and bias—not to mention its staggering energy needs. *See, e.g.* Öykü Isik, Amit Joshi & Lazaros Goutas, *4 Types of Generative AI Risk and How to Mitigate Them*, Harv. Bus. Rev. (May 31, 2024), https://hbr.org/2024/05/4-types-of-gen-ai-risk-and-how-to-mitigate-them (reviewing and categorizing various risks of generative AI); Kate Crawford, *Generative AI's environmental costs are soaring—and mostly secret*, Nature (Feb. 20, 2024), https://www.nature.com/articles/d41586-024-00478-x ("Within years, large AI systems are likely to need as much energy as entire nations.").

[202] Interpreting the *Oracle* decision, Ginsburg aptly observes that if "*verbatim* copying 'to create new products' were deemed 'transformative' in general, it would be difficult to imagine what kind of copying, short of outright piracy of the entire work, would not be transformative." Ginsburg, *supra* note 155, at 7; *see also Harper & Row*, 539 U.S. at 569 ("Any copyright infringer may claim to benefit the public by increasing public access to the copyrighted work.").

[203] *See infra* notes 212-33 and accompanying text.

[204] *Warhol*, 598 U.S. at 532-33.

[205] *See supra* notes 33 and 114 and accompanying text (AI systems must train on millions of works); *infra* note 223 and accompanying text (licensing would be impossible).

[206] *See* OpenAI, *Pricing*, https://openai.com/chatgpt/pricing/ (last visited May 15, 2024) (listing subscription fees for OpenAI's ChatGPT products); Stability AI, *Stability AI Membership*, https://stability.ai/membership (last visited May 15, 2024) (listing membership fees for Stability AI

commercially marketed AI system is by definition commercial in nature.[207] Although not in itself dispositive, this factor weighs significantly against fair use.[208]

2. *AI companies copy highly creative works*

With respect to the second factor, the "nature of the works" at issue, AI companies acknowledge that they have copied countless expressive works—such as books, movies and visual art—that lie at the heart of copyright.[209] This presumptively weighs against fair use.[210]

3. *AI companies copy works in their entirety*

Factor three considers the amount of the work that was copied. There is no dispute that AI companies reproduce copyrighted works in their entirety to assemble training sets. Further, as explained above, during the training process, each work is broken down into small segments and algorithmically encoded into the model. The core activities of AI development, then, involve the reproduction of entire works.

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products); Anthropic, *Meet Claude*, https://www.anthropic.com/claude (last visited May 15, 2024) (listing subscription fees for Anthropic's Claude products).

[207] This holds true even though, in some cases, AI companies acquire training materials assembled by academic and other nonprofit researchers. *See Andy Baio*, *AI Data Laundering: How Academic and Nonprofit Researchers Shield Tech Companies from Accountability*, Waxy (Sept. 30, 2022), https://waxy.org/2022/09/ai-data-laundering-how-academic-and-nonprofit-researchers-shield-tech-companies-from-accountability// (discussing common practice of "technology companies working with AI to commercially use datasets and models collected and trained by non-commercial research entities like universities or non-profits"). Even if a researcher's activities are conducted under the auspices of a nonprofit institution, this does not negate the commercial purpose of the follow-on AI entity. Nor does it negate a finding of for-profit use on the part of the researcher. In *Weissman v. Freeman*, 868 F.2d 1313 (2d Cir. 1989), for example, the Second Circuit determined that an academic's copying of a scientific paper, though not for monetary gain, was nonetheless a for-profit activity for purposes of the first fair use factor. *Id.* at 1324 ("Particularly in an academic setting, profit is ill-measured in dollars. Instead, what is valuable is recognition because it so often influences professional advancement and academic tenure."); *see also Harper & Row*, 471 U.S. at 562 ("The crux of the profit/nonprofit distinction is not whether the sole motive of the use is monetary gain but whether the user stands to profit from exploitation of the copyrighted material without paying the customary price."); *Worldwide Church of God v. Phila. Church of God, Inc.*, 227 F.3d 1110, 1117 (9th Cir. 2002) (same); *Am. Geophysical Union v. Texaco Inc.*, 60 F.3d 913, 914-15, 921-22, 931 (2d Cir. 1994) ("*Texaco*") (photocopying of journal articles by Texaco scientists for internal research purposes held not a fair use in part because Texaco reaped indirect economic advantage and "avoid[ed] having to pay at least some price to copyright holders").

[208] *See Campbell*, 510 U.S. at 585 (commercial use is a factor that "'tends to weigh against fair use'") (quoting *Harper & Row*, 471 U.S. at 562)).

[209] *Id.* at 586 ("This factor calls for recognition that some works are closer to the core of intended copyright protection than others, with the consequence that fair use is more difficult to establish when the former works are copied.").

[210] *Id.*; *Sony Corp. v. Universal City Studios, Inc.*, 464 U.S. 417, 455 n.40 (1984) (copying a news broadcast may have different fair use implications than copying a motion picture).

On the output end, as discussed above, works from training materials may be replicated in all or in part in AI productions. Further, in canvassing the internet for additional material to enhance output, RAG technology appropriates the entirety or significant portions of the external sources it gathers.

In short, factor three weighs heavily against fair use, especially since the works are being used for their expressive content rather than functional purposes.[211]

4. *AI copying harms the market for the copied works*

The final fair use factor concerns the effect of the use upon the potential market for or value of the copyrighted work. In the AI context, this factor encompasses at least three different concerns: harm to the market for the original works based on generation of competing copies thereof; harm to the market for licensed derivatives based on those works; and harm to the licensing market for training use of the works.

A close copy of a work that can serve as a substitute for the original invades the market for the original. Likewise, a derivative based on a copied work may supplant or interfere with licensing opportunities for adaptations of the original. Especially in a commercial context, both types of output weigh against fair use.[212]

Creators and owners of copyrighted works are deeply concerned that generative AI systems that copy their works in the training process have the capacity to generate products that, even if not overt reproductions, are stylistically similar enough to the original works to serve as competing substitutes.[213] Imitative content may incorporate elements of a copyrighted original or originals

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[211] *Compare Google Books*, 804 F.3d at 221-22 ("While Google makes an unauthorized digital copy of the entire book, it does not reveal that digital copy to the public …. Google [thus] satisfies the third factor test.") *with TVEyes*, 883 F.3d at 179 ("Th[e] [third] factor clearly favors Fox because TVEyes makes available virtually the entirety of the Fox programming that TVEyes users want to see and hear.").

[212] *See Warhol*, 598 U.S. at 536 (where secondary work is both substitutional and used for commercial purposes, this "counsels against fair use, absent some other justification"); *Campbell*, 510 U.S. at 590 (fair use analysis must consider not only harm to original work but market for derivatives).

[213] For example, as articulated in the record labels' suit against AI music generator Udio:

> The capacity for a generative AI service to produce convincing imitations of genuine sound recordings starts with copying a vast range of sound recordings. When those who develop such a service steal copyrighted sound recordings, the service's synthetic musical outputs could saturate the market with machine-generated content that will directly compete with, cheapen, and ultimately drown out the genuine sound recordings on which the service is built.

Compl. ¶ 4, *UMG Recordings, Inc. v. Uncharted Labs, Inc.*, No. 1:24-cv-04777 (S.D.N.Y. June 24, 2024) ("*UMG* Compl."); *see also* Kashmir Hill, *This Tool Could Protect Artist from A.I.-Generated Art That Steals Their Style*, N.Y. Times (Feb. 13, 2023), that render the imitation an infringing derivative.[214] In other instances, however, generative output may not meet the traditional test for substantial similarity.[215] Even if not technically infringing, output that is recognizable as derived from a particular creator's works should weigh against a claim that the AI model's copying of such works during training was a fair use. That is, even if infringement cannot be established based on generated output, output that obviously imitates an artist's distinctive style may establish market harm resulting from copying at the training stage because that copying yielded a competing substitute.[216]

AI companies rely on unauthorized copying—largely accomplished by scraping the internet—to develop their models. Most creators and copyright owners became aware of the appropriation of their works by AI companies only after generative AI burst into public view in 2022. One consequence of this revelation has been a flood of lawsuits filed by writers, visual artists, musicians and others asserting copyright infringement claims and additional causes of action.[217] A second consequence is a quickly developing market for licensing of copyrighted content to AI companies for training and operation of their systems.

Since 2023, OpenAI, the company behind ChatGPT, has entered into content licensing deals with Axel Springer, the publisher of POLITICO and Business Insider; News Corp, which owns the Wall Street Journal, the New York Post, the Times and the Sunday Times; Dotdash Meredith, a large publisher of online content; and the Associated Press, to name just some.[218] Google

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https://www.nytimes.com/2023/02/13/technology/ai-art-generator-lensa-stable-diffusion.html (discussing concerns of visual artists whose work is being imitated by generative AI).

[214] *See supra* note 79 and accompanying text (discussing derivative output).

[215] *See* Sobel, *Elements*, at 38-41 (explaining why it may be difficult to establish substantial similarity in artistic style).

[216] The production of imitative content may also be probative of copying at the training stage.

[217] *See, e.g.*, *Andersen v. Stability AI, Ltd.*, No. 3:23-cv-00201 (N.D. Cal. filed Jan. 13, 2023) (visual art); *Tremblay v. OpenAI, Inc.*, No. 3:23-cv-03223 (N.D. Cal. filed June 28, 2023) (books) and *Silverman v. OpenAI, Inc.*, No. 3:23-cv-03416 (N.D. Cal. filed July 7, 2023) (books) (*Tremblay* and *Silverman* now consolidated in *In re OpenAI ChatGPT Litig.*, No. 23-cv-03223); *Concord Music Group, Inc. v. Anthropic PBC*, No. 3:23-cv-01092 (M.D. Tenn. filed Oct. 18, 2023) (song lyrics); *New York Times v. Microsoft Corp*., No. 1:23-cv-11195 (S.D.N.Y. filed Dec. 27, 2023) (news articles); *UMG Recordings, Inc. v. Suno, Inc.*, No. 1:24-cv-11611 (D. Mass. filed June 24, 2024) (sound recordings); *UMG Recordings, Inc. v. Uncharted Labs,* No. 1:24-cv-04777 (S.D.N.Y. filed June 24, 2024) (sound recordings).

[218] Angela Cullen & Jackie Davalos, *OpenAI to Pay Axel Springer Tens of Millions to Use News Content*, Bloomberg Law (Dec. 13, 2023), https://news.bloomberglaw.com/tech-and-telecom-law/openai-to-pay-axel-springer-tens-of-millions-to-use-news-content; Guardian staff and agencies, *Open AI and Wall Street Journal owner News Corp sign content deal*, The Guardian (May 22, 2024), https://www.theguardian.com/technology/article/2024/may/22/openai-chatgpt-news-corp-deal; Sara Fischer, *OpenAI inks licensing deal with Dotdash Meredith*, Axios (May 7, 2024), https://www.axios.com/2024/05/07/openai-dotdash-meredith-licensing-deal; Matt O'Brien*, ChatGPT-maker OpenAI signs deal with AP to license news stories*, AP (July 13, 2023), https://apnews.com/article/openai-chatgpt-associated-press-ap-f86f84c5bcc2f3b98074b38521f5f75a.

reached an agreement with Reddit to use Reddit data to train its AI models.[219] Universal Music has entered into a partnership with AI technology company SoundLabs to provide an "ethically" trained voice cloning tool for its artists.[220] Disney Music has licensed AI music startup AudioShake to "open up" Disney's historic catalog of works to new uses.[221] The Copyright Clearance Center, an organization that licenses journal articles and other text-based materials, has extended its collective licensing service to cover AI uses.[222] These are just some of the licensing arrangements that have been publicly disclosed; undoubtedly there are many deals that have not been made public or are still in the pipeline.

The narrative being promoted by AI companies and their defenders is that licensing content to train and develop AI systems is "impossible."[223] But AI companies have demonstrated that they are capable of entering into license arrangements when they see value in the licensed content. We are still in the early days of generative AI licensing, so the market is not yet mature. But it is active.[224] In addition to licensing deals for large corpora of works, we will likely see growth in niche markets for specialized content and works of individual creators who choose to participate.

AI advocates have been known to assert that the appropriation of copyrighted works to develop and operate AI models does not interfere with copyright owners' legitimate economic interests because the authors of books, movies and music did not produce those works with the intent of populating generative models.[225] While such an assertion may be true as far as it goes, the claim

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[219] Annelise Gilbert, *Google-Reddit AI Deal Hearlds New Era in Social Media Licensing*, Bloomberg Law (Mar. 7, 2024), https://news.bloomberglaw.com/ip-law/google-reddit-ai-deal-just-the-start-for-social-media-licensing,

[220] *See* Mandy Dalugdug, *Universal Music artists get access to AI voice cloning tool via UMG's new deal with tech startup SoundLabs*, Music Business Worldwide (June 19, 2024), https://www.musicbusinessworldwide.com/universal-music-artists-get-access-to-ai-voice-cloning-tool-via-umgs-new-deal-with-tech-startup-soundlabs/.

[221] Murray Stassen, *Disney Music Group Strikes Deal With AI Music Startup AudioShake to 'Unlock New Listening and Fan Engagement Experiences' for Its Catalog*, Music Business Worldwide (July 15, 2024), https://www.musicbusinessworldwide.com/disney-music-group-strikes-deal-with-ai-music-startup-audioshake-to-unlock-new-listening-and-fan-engagement-experiences-for-its-catalog/.

[222] Ed Nawotka, *CCC Launches Collective Licensing for AI*, Publishers Weekly (July 16, 2024), https://www.publishersweekly.com/pw/by-topic/digital/copyright/article/95512-ccc-launches-collective-licensing-for-ai.html.

[223] *See* Damle, *supra* note 2, at 3, 12-16 ("[E]veryone agrees that it is impossible for AI developers to negotiate and acquire licenses from every rightsholder who owns a[] copyright interest in the data used to train AI models."); Lemley & Casey, *supra* note 7, at 770 ("[G]iven the large number of works an AI training data set needs to use and the fact that thousands, if not millions, of different people own those works, AI companies can't simply license all the underlying photographs or text …."); *see also* R Street, *supra* note 75 ("A system that required follow-on creators to negotiate with and pay those they learned from would inhibit, rather than promote, the very artistic progress our IP laws seek to encourage.").

[224] Pointing to the robust market for "immensely valuable" user data, Sobel asserted in 2017 that "there is already a thriving market for the data that fuel expressive machine learning." Sobel, *Fair Use*, *supra* note 14, at 76.

[225] *See, e.g.*, Lemley & Casey, *supra* note 7, at 776 ("The copyright owner of a book or photograph doesn't create that work in hopes of selling it to AIs.").

is unconvincing. Authors and artists of just a few decades ago likely did not anticipate that their works would be accessed and consumed through mobile phones or watches. But no doubt they expected their copyrights to continue to protect the expressive content embodied in those works even if that content was exploited through means yet to be known.

In evaluating market impact, it is critical to assess not only existing modes of exploitation, but future markets as well. As the Supreme Court has emphasized, the statutory mandate is to consider "'the effect of the use upon the *potential* market for or value of the copyrighted work'"[226]—not just the current market—including opportunities "that creators of original works would in general develop or license others to develop."[227] This includes potential markets for derivative uses.[228] In other words, any potential market, including a nascent or still-developing market, is to be considered. In *American Geophysical Union v. Texaco Inc.*,[229] for instance, the Second Circuit rejected Texaco's fair use defense to unlicensed copying of journal articles for internal research purposes when licenses were available from the copyright owners.[230] The court determined that publishers' lost licensing revenues would result in "substantial harm to the value of their copyrights."[231]

As the Second Circuit has repeatedly affirmed, "'[i]t is indisputable that, as a general matter, a copyright holder is entitled to demand a royalty for licensing others to use its copyrighted work, and that the impact on potential licensing revenues is a proper subject for consideration in assessing the fourth factor.'"[232] The licensing market for AI training materials is already far from hypothetical. The fact that AI companies are commercially motivated entities, many with significant economic resources, points to continued growth in this area.[233] The existence of a rapidly evolving market for materials to build and operate AI systems weighs powerfully against a finding of fair use that could extinguish that market.

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[226] *Campbell*, 510 U.S. at 590 (quoting 17 U.S.C. § 107 (emphasis added)); *Harper & Row*, 471 U.S. at 566 (same).
[227] *Campbell*, 510 U.S. at 592; *see also Seuss*, 983 F.3d at 460 (same).
[228] *Id.* at 590 (fair use analysis "'must take account not only of the harm to the original but also of harm to the market for derivative works'") (quoting *Harper & Row*, 471 U.S. at 568)).
[229] 60 F.3d 913.
[230] *Id.* at 930-31 ("Though the publishers still have not established a conventional market for the direct sale and distribution of individual articles, they have created, primarily through the C[opyright] C[learance] C[enter], a workable market for institutional users to obtain licenses for the right to produce their own copies of individual articles via photocopying.").
[231] *Id.* at 931.
[232] *TVEyes*, 883 F.3d at 180 (quoting *Bill Graham Archives v. Dorling Kindersley Ltd.*, 448 F.3d 605, 624 (2d Cir. 2006)); *see also Texaco*, 60 F.3d at 929 (same).
[233] *See* Lemley & Casey, *supra* note 7, at 765 ("Commerciality often goes hand in hand with a market effect…. M[achine] L[earning] companies might be natural candidates for a licensing market: large for-profit companies that stand to benefit financially from using copyrighted works ….").

Conclusion

The Copyright Act protects works of human authors, not of machines.[234] At this early stage, we do not yet know how generative AI will impact human authorship, or creative culture in general. Will there be less incentive for humans to create works, and for publishers to invest in and disseminate those works, because the human works are competing with AI-generated content? Conversely, will humans find it worthwhile to spend time engaging with AI systems to produce content that is not protected by copyright and can be freely exploited by others? If human authorship declines, will there be a corresponding decline in the appeal of AI-generated content as AI machines rely on the same materials over and over? Is the ability to produce an infinite number of texts, images or songs unconnected with a human artist actually of meaningful social value?[235] Is an essential aspect of the human experience of and appreciation for art the fact that it is made by a human author?[236]

This article explains why unauthorized copying by AI companies to build and operate generative AI systems is not, as claimed, "quintessential fair use." There is no fair use precedent that legitimizes mass copying and exploitation of the expressive content of creative works by for-profit entities. Apart from doctrinal concerns, an overly broad application of fair use to exempt unconstrained copying by AI companies could effect a potentially enormous transfer of value from the creators and owners of copyrighted works to the commercial entities that seek to exploit

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[234] As confirmed in *Thaler v. Perlmutter*, 687 F. Supp. 3d 140, 147-48 (D.D.C. 2023), which upheld the Copyright Office's refusal to register an AI-generated work, in referring to "authors" the Copyright Act means human authors. *Id.* at 147. As the *Thaler* court explained, although copyright "is designed to adapt with the times," there has been "a consistent understanding that human creativity is the *sine qua non* at the core of copyrightability, even as that human creativity is channeled through new tools or into new media." *Id.* at 146.

[235] *See, e.g.*, *UMG* Compl., *supra* note 213, ¶ 12 (alleging that AI defendant Udio generates 10 music files per second, or 6 million files per week, from copyrighted sound recordings).

[236] Some more broadly question whether the enormous investment in generative AI will yield net social benefits. As encapsulated by MIT professor Daron Acemoglu in an AI-focused report by Goldman Sachs:

> Technology that has the potential to provide good information can also provide bad information and be misused for nefarious purposes. I am not overly concerned about deepfakes at this point, but they are the tip of the iceberg in terms of how bad actors could misuse generative Al. And a trillion dollars of investment in deepfakes would add a trillion dollars to GDP, but I don't think people would be happy about that or benefit from it.

Nathan, Grimberg & Rhodes, *Gen AI: Too Much Spend, Too Little Benefit*, Goldman Sachs, at 5 (June 25, 2025) (interview of Daron Acemoglu); *id.* at 10 ("AI technology is exceptionally expensive, and to justify those costs, the technology must be able to solve complex problems, which it isn't designed to do.") (interview of Jim Covello, head of Goldman Sachs global equity research).

them. An unprecedented exception to copyright with such far-reaching consequences is not a question of fair use but rather a fundamental question of copyright policy for Congress to decide.[237]

[237] "The Congress shall have Power … To promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries." Const. art. I, § 8, cl.8. In fact, Congress has already begun to consider the challenges presented by generative AI. *See, e.g.*, *Artificial Intelligence and Intellectual Property: Part II – Identity in the Age of AI: Hearing Before the U.S. H. Comm. on the Judiciary, Subcommittee on Courts, Intellectual Property, and the Internet*, 118th Cong. (Feb. 2, 2024); *Oversight of A.I.: Legislating on Artificial Intelligence: Hearing Before the U.S. S. Comm. on the Judiciary*, 118th Cong. (Sept. 12, 2023); *Artificial Intelligence and Intellectual Property: Part I–Interoperability of AI and Copyright Law: Hearing Before the U.S. H. Comm. on the Judiciary, Subcommittee on Courts, Intellectual Property, and the Internet*, 118th Cong. (May 17, 2023). In examining these issues, Congress has the benefit of advice from the U.S. Copyright Office, which has undertaken a multipart study on copyright and artificial intelligence. *See* U.S. Copyright Office, *Artificial Intelligence Study*, https://www.copyright.gov/policy/artificial-intelligence/ (last visited July 8, 2024) ("Copyright Office AI Study"). The Copyright Office issued the first installment of its study in July 2024, recommending legislation to address AI-generated "digital replicas," or deepfakes, that imitate individuals' images or voices. U.S. Copyright Office*, Copyright and Artificial Intelligence: Part 1: Digital Replicas* 57 (2024), https://www.copyright.gov/ai/Copyright-and-Artificial-Intelligence-Part-1-Digital-Replicas-Report.pdf ("The Copyright Office agrees with the numerous commenters that have asserted an urgent need for new protection at the federal level.") A bipartisan bill to protect against deepfakes was introduced by the Senate on July 31, 2024. *See* Nurture Originals, Foster Arts, and Keep Entertainment Safe (NO FAKES) Act of 2024, S. 4875, 118th Cong. (2024).